PDF EXPORT

CAESAR II Users Guide

Octave Docs

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Static Output Processor

Main window ribbon: **Home** > **Reports** > **Static**

Main window ribbon: **Output** > **Reports** > **Static**

In the **Classic Piping Input** window:

CAESAR II Tools toolbar: **View Static Results**

Provides an interactive view of static analysis results.

The classic **Static Output Processor** displays analysis results in a tabular form, in a graphical animated form, or a combination of the two. Use commands in the **Static Output Processor** to:

- Interactively review reports for any selected combination of load cases and/or report types.
- Print or save to file copies for any combination of load cases and/or report types.
- Add title lines to output reports.
- Select extended or summarized versions of most standard reports.

TIP This command can be manually run at any time when the default **New Analysis Reviewer** is selected for Preferred Output Reviewer in the Configuration Editor By default, results are displayed and customized in the New Analysis Reviewer.

Load Cases Analyzed

List the load cases analyzed in the current job.

Standard Reports

Lists the available reports associated with those load cases. See Work with Reports.

General Computed Results

Lists reports, such as input listings or hanger selection reports, that are not associated with load cases.

Custom Reports

Lists generated or imported custom reports. See Work with Reports and Report Template Editor.

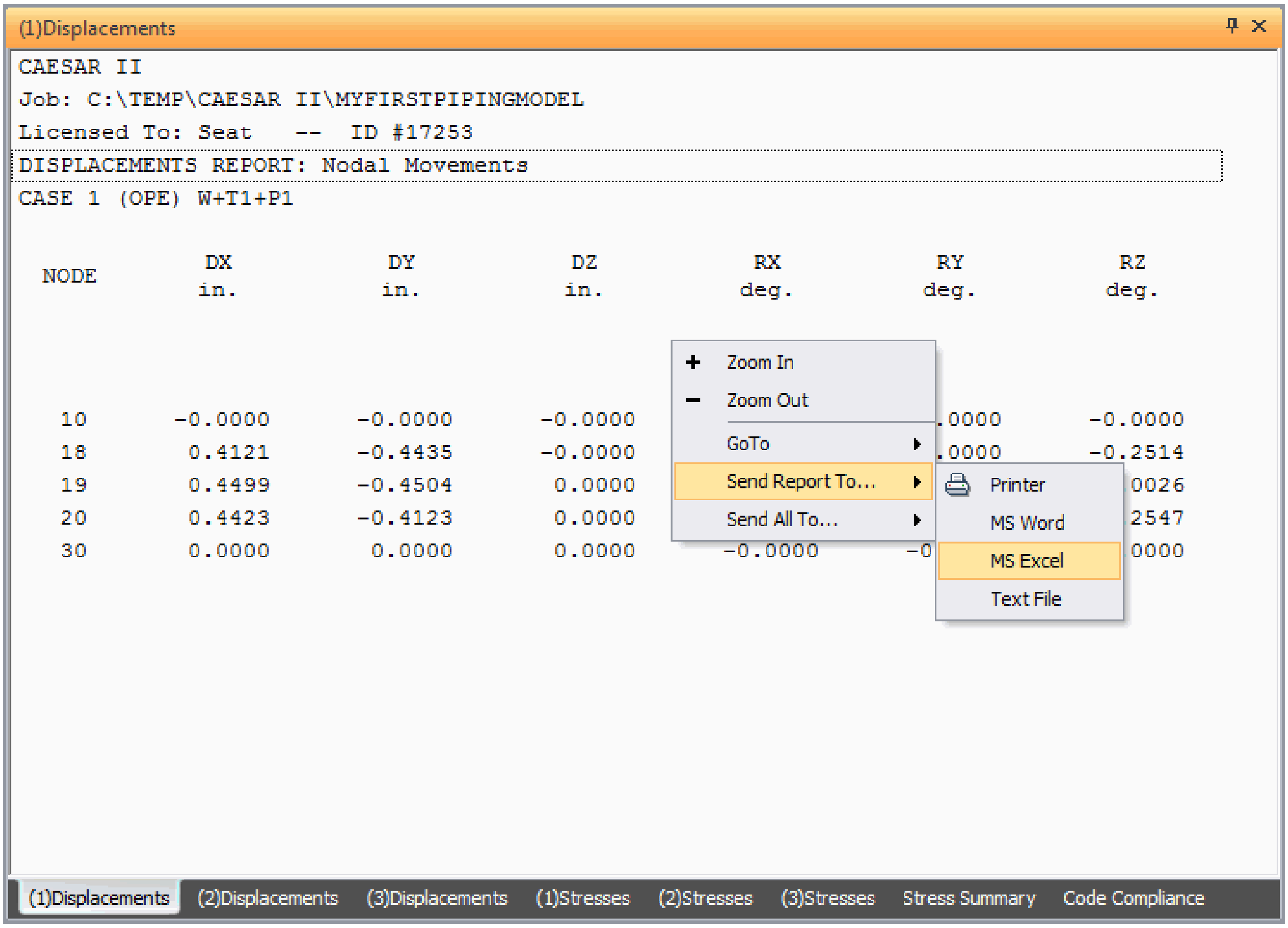
Output Viewer Wizard

Selects specific reports and reviews their order before sending the output to the selected device.

Work with Reports

When using the **Send to Screen** option on the Output Viewer Wizard, reports display in a tabbed **Reports Viewer** window.

Detach individual reports from the tabbed view and position them around the screen. Additionally, dock a report next to other opened reports for a comparison view. Select the tab at the bottom of the report, and while holding down the mouse, move the report. The outline shadow shows the new location of the report. Release the mouse button to place the report in the new location.



To change the company text displayed in the **Licensed To** field, edit the first line of Company.txt, found in the System Folder. The maximum length is 64 characters.

When a report is open, double-click the column headings to sort the report by ascending or descending value order. Column order can be re-arranged by dragging columns to another location. Also adjust the column size or hide the column altogether. All changes are for the current report in the current viewing session. To make permanent changes to the report, use the Report Template Editor.

Print or save individual reports to a text file, to Microsoft Word, or to Microsoft Excel by selecting **Send Report To** or **Send All To** from the right-click menu.

While the report is active, adjust the display properties available from the **View** menu, change the background color, and turn on horizontal and vertical grid lines. Grid lines can help in generating better print results.

Select **View > Change Page Break** to adjust the page configuration for an active report. Also scale the report to fit on one page or adjust it to fit on a specified number of pages by using the **Allow Adjustment of Page Breaks** and **Show Page Break Lines** options.

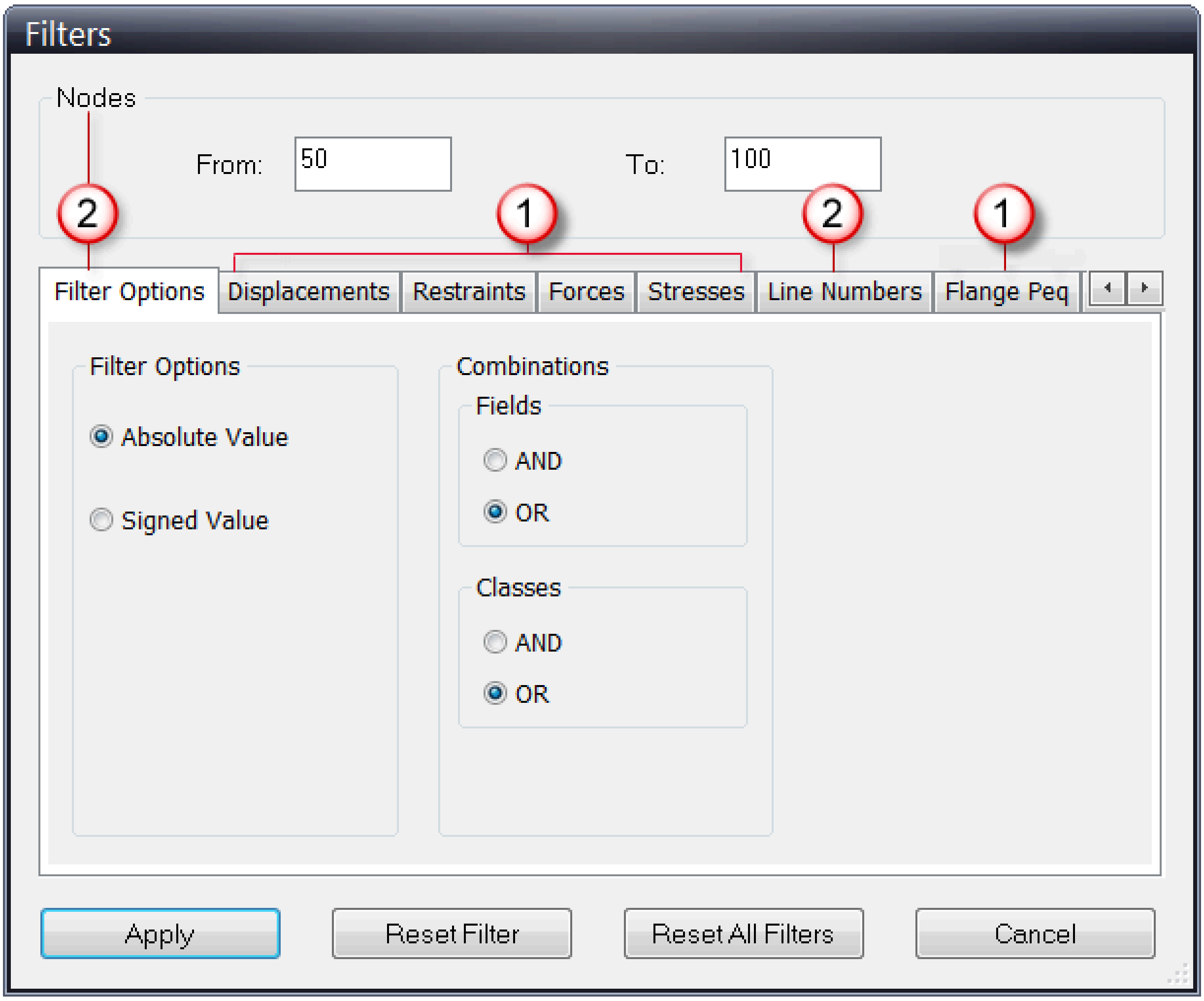
Filter Reports

Static Output Processor menu: **Filters**

Allows you to filter your static output reports. Filtering customizes output to just the information you want to see.

Some filters apply globally to output reports (meaning any kind of report you generate from the **Output Processor**) or apply to a specific report type (such as only to restraint reports).

Filters Dialog



1 Report-Specific Filters
2 Global Filters

Report-Specific Filters

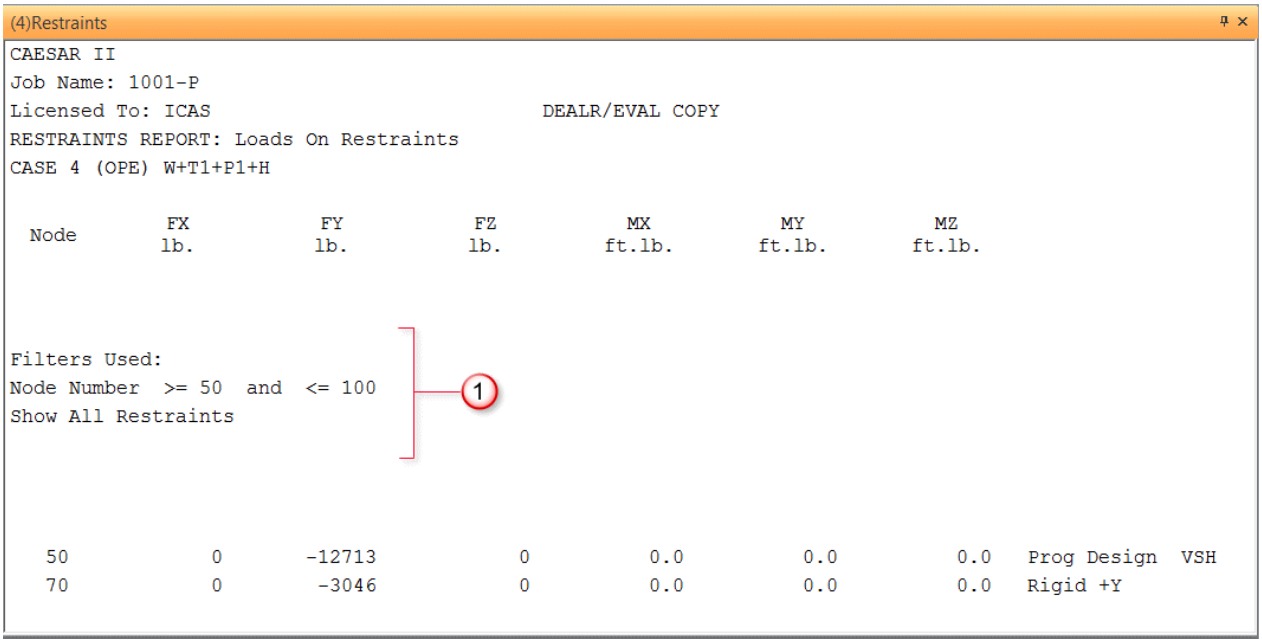
Apply report-specific filters when you want the software to filter only reports that are related to the filter setting. For example, if you wanted to generate a report to see all the allowable stresses in a piping system that are greater than 80 percent, set the **Percent** box on the **Stresses** tab to **>80**. Then, when you generate any of the stress-related reports in the **Output Processor**, the software filters the data to show only those stresses greater than 80 percent.

Global Filters

Apply global filters to node number or line number ranges for the software to filter all reports that include the node or line numbers specified. For example, if you filter on node numbers from 10 to 100, then for any report you generate in the **Output Processor**, the software filters and shows only the data that is applicable to nodes 10 to 100.

NOTES

- Filters do not apply to the summary information that appears at the top of a report.
- The software shows the filters applied near the top of the output report, as shown below.



```
(4)Restraints
CAESAR II
Job Name: 1001-P
Licensed To: ICAS                              DEALR/EVAL COPY
RESTRAINTS REPORT: Loads On Restraints
CASE 4 (OPE) W+T1+P1+H

 Node        FX          FY          FZ          MX          MY          MZ
             lb.         lb.         lb.       ft.lb.      ft.lb.      ft.lb.


Filters Used:
Node Number  >= 50  and  <= 100
Show All Restraints



  50          0       -12713          0         0.0         0.0         0.0  Prog Design  VSH
  70          0        -3046          0         0.0         0.0         0.0  Rigid +Y
```

1 Output Reports Show Filters Used

To filter reports

1. Select **Filters** on the **Output Processor** menu.
2. Choose any global filter options. Global filters apply to all reports generated in the **Output Processor**. For more information on the global filters, see Filter Options Tab.
   a. To apply a global filter based on node numbers, type the **From** node and **To** node numbers.
   b. To select other global options for filters, select the **Filter Options** tab.
3. Choose report-specific filter details on each of the class tabs. Each tab contains related fields with a drop box and an edit box. Each corresponding edit box displays the value in which to compare.

   Each of the drop boxes has a list of comparison operators:

Operator	Description
>	Greater than
>=	Greater or Equal
<	Less than
<=	Less or Equal
==	Equal
\=	Not Equal

4. Select **Apply** to define the filter.

Filters Dialog

Specifies report results to display only needed information. Certain filters apply to output reports globally (meaning any kind of report you generate from the Static Output Processor) while others apply to a specific report type (such as only to the Restraints Report). See Filter Reports.

Filter Options Tab

Displacements Tab

Restraints Tab

Forces Tab

Stresses Tab

Line Numbers Tab

Flange Peq Tab

Flange NC-3658.3 Tab

Filter Options Tab

Select filter options that the software applies to all reports generated in the **Output Processor**. Set filter options from the **Filter Options** tab on the **Filters** dialog, accessed by selecting **Filters** in the Output Processor menu. Specify filter settings, such as setting the signed value or the combination of fields filtered, which the software applies globally to all reports you generate.

From Node

Enter both a **From** and **To** node number and the software filters based on nodes in the specified range display. In other words, only one of the node pair must fall within the specified range. If you enter just a **From** node number, then the software displays the **From** node and all nodes greater than it. The software filters and displays all node pairings that have at least one node (**To** or **From**) in the filter setting for the node range.

To Node

Enter both a **From** and **To** node number and the software filters based on nodes in the specified range display. If you enter just a **To** node number, then the software displays the **To** node and all nodes less than it. The software filters and displays all node pairings that have at least one node (**To** or **From**) in the filter setting for the node range.

Filter Options

Select the appropriate filter option for values: **Absolute Value** or **Signed Value**. The software defaults to filtering by the magnitude, regardless of the sign or direction. Filter by a specific direction of load or displacement. This feature is particularly useful when looking for lifting off the supports in directional restraints (such as +Y).

Combinations (Fields or Classes)

Select the appropriate filter option for field or class combinations. Fields refer to the data inputs in each tab class. Classes refers to the major types of output, for example **Displacements**, **Restraints**, **Forces**, or **Stresses**. Classes have separate tabs in the **Filters** dialog. For example, **DX** and **RZ** are fields in the **Displacements** class, **FX** and **MZ** are fields in the **Restraints** class, and **Code Stress** and **Bending Stress** are fields in the **Stresses** class.

Reset Filter

Select to reset the filter settings for the currently-active tab.

Reset All Filters

Select to reset all the filter settings for all filters, including the options and the report-specific filter classes.

Displacements Tab

Specifies filter criteria for all reports relating to displacements. Use the operator boxes to specify comparison operators for the filter values.

For example, to look at any place in the model where the pipe is lifting off the support, set the **Displacements** filter class to a positive value of greater than one inch in the Y-axis direction (**DY**) to show which pipes have lifted off the support vertically by more than an inch.

When you run the **Displacements** output report, the software shows only those pipes in your model that have been displaced by more than an inch. Also set the **Filter Options Classes** to **AND**, and then generate the **Restraints Summary Extended** report to show all restraints with a displacement off the pipe of one inch or more.

For more information on filtering output reports, see Filters Dialog and Filter Reports.

NOTE The software does not apply report-specific filter criteria to custom reports that use combination classes.

Restraints Tab

Specifies filter criteria for all reports relating to restraints. Use the operator boxes to specify comparison operators for the filter values.

For example, if you set the **Restraints** filter class to show all restraints with forces greater than 10,000 lbs on the y-axis (**FY**), then the software shows only restraints that have **FY** values greater than 10,0000 lbs on the **Restraints**, **Local Restraints**, or **Restraints Summary** output reports.

Also select one of the **Include** radio buttons to filter and include all restraints in your results, include all restraints that do not have CNodes (**None with CNodes**), or include all restraints with only CNodes (**Only with CNodes**).

For example, select to include **None with CNodes**, if you want to filter out internal load on a CNoded restraint. This lets you see the total dead weight of a pipe model.

Select **Bi-Linear** for buried pipe models that also go above ground. In most cases, you do not need to select this filter option.

For more information on filtering output reports, see Filters Dialog and Filter Reports.

NOTE The software does not apply report-specific filter criteria to custom reports that use combination classes.

Forces Tab

Specifies filter criteria that the software applies to all reports relating to force. Use the operator boxes to specify comparison operators for the filter values.

For example, if you set the **Forces** filter class to all forces in the Y-axis direction greater than 10,000 pounds, the software shows only y-axis forces (**FY**) greater than 10,000 pounds on the **Local Element Forces** and **Global Element Forces** output reports.

For more information on filtering output reports, see [Filters Dialog Box] and [Filter Reports].

NOTE The software does not apply report-specific filter criteria to custom reports that use combination classes. This filter applies only to internal forces and moments.

Stresses Tab

Specifies filter criteria for the various stress output reports (**Stresses**, **Stresses Extended**, and **Stress Summary**). Set up filter criteria based on Axial, Bending, Torsion, Hoop, Max 3D, Code and Allowable stress factors in combination with the magnitude. In addition, filter stress data on the reports based on SIFs (in- and out-of-plane), and on a percentage of stress. Use the operator boxes to specify comparison operators for the filter values.

For example, if you set the **Stresses** filter class **Percent** box to **>70**, the software filters the stress-related reports to show only elements having greater than a 70 percent stress.

For more information on filtering output reports, see [Filters Dialog] and [Filter Reports].

NOTE The software does not apply report-specific filter criteria to custom reports that use combination classes.

Line Numbers Tab

Assigns filter criteria based on line numbers that the software applies globally for all output reports. If the model has assigned line numbers, set up filter criteria based on those numbers to look at output results for only certain parts of the model.

For example, apply line number ranges for the software to filter reports to only include elements that belong to a line number. For example, if you filter based upon line numbers, then for any report you generate in the **Output Processor**, the software filters and shows only the data that includes information on systems with the specified line numbers. When the software shows the **Line Number** box as **Unassigned**, it means the model did not have any line numbers assigned and cannot be filtered based on that information.

For more information on filtering output reports, see [Filters Dialog] and [Filter Reports].

Flange Peq Tab

Specifies filter criteria for flange output reports related to the Kellog Equivalent Pressure Method (Peq). Set up filter criteria based on **Axial Force**, **Bending Moment**, **Gasket Diameter**, **PEquivalent**, **Rating**

Temperature, **Allowable Pressur**e, and a **Ratio** factor. Use the operator boxes to specify comparison operators for the filter values.

For example, if you specify the Flange Peq filter class to filter based on a Ratio of 20 percent, then the software filters output reports to only show equivalent pressures for flanges that are 20 percent of the maximum rated pressure. This information indicates how close you are to the edge of the flange.

For more information on filtering output reports, see Filters Dialog and Filter Reports.

NOTE The software does not apply report-specific filter criteria to custom reports that use combination classes.

Flange NC-3658.3 Tab

Specifies filter criteria for flange output reports related to ASME B&PVC Section III Subsection NC-3658.3 Method (NC-3658.3). Set up filter criteria for all NC-3658.3 reports based on **Torsion Moment**, **Bending Moment**, **Bolt Circle Diameter**, **Bolt Area**, **Flange Stress**, **Allowable Stress**, and a **Ratio** factor. Use the operator boxes to specify comparison operators for the filter values.

For example, if you specify for the Flange NC-3658.3 filter class to with a Ratio of 30 percent or greater, then the software reports on only flanges of the NC method that are 30 percent of the maximum rated equivalent pressure for the flange.

For more information on filtering output reports, see Filters Dialog and Filter Reports.

NOTE The software does not apply report-specific filter criteria to custom reports that use combination classes.

Print or Save Reports to File Notes

The tabular results brought to the screen may be sent directly to a printer. Different combinations of load cases and report types may be chosen, each followed by the **File-Print** command, to create a single report.

	Prints copies of the reports. To print copies of multiple reports as a single report, use the Output Viewer Wizard to populate the report order tree, select **Send To Printer** and then **Finish**.
	Sends reports to a file (in ASCII format) rather than the printer. After selection, a dialog displays where you select the file name. To change the file name for a new report, select **File-Save As**.

Typically, the set of output reports to print out for documentation purposes is:

Load Case	Report	Purpose
SUSTAINED	STRESS	Code compliance
EXPANSION	STRESS	Code compliance
OPERATING	DISPLACEMENTS	Interference checks
OPERATING	RESTRAINTS	Hot restraint, equipment loads
SUSTAINED	RESTRAINTS	As-installed restraint, equipment loads

NOTE Load cases used for hanger sizing produce no reports. Also, the hanger table and hanger table with text reports are printed only once even though more than one active load case may be highlighted.

To save multiple reports as a single report to a file, use the **Output Viewer** Wizard.

NOTES

- The signs in all CAESAR II reports show the forces and moments that act "ON" something. The **Element Force/Moment** report shows the forces and moments that act "ON" each element to keep that element in static equilibrium. The **Restraint Force/Moment** report shows the forces and moments that act "ON" each restraint.
- When sending reports to MSWord, if a file named header.doc exists in the CAESAR II System folder, its contents are read and used as the page header when CAESAR II exports the report to Microsoft Word. The intent is that header.doc contains the company logo, address details and formatting for tables. The interface uses a style names report table which is set up in header.doc.

Load Cases Analyzed

Lists all load cases which have been analyzed for the current job. The cases are numbered and labeled with the type (load category) addressed by the case.

The load case description also includes the individual load components that contributed to the load case.

The results for a load case can be viewed by selecting the load case. Multiple load cases can be selected using the SHIFT and CTRL keys in combination with the mouse.

Load types are:

OPE

Operating case. For B31.1 and B31.3 (and similar codes) this case is not a code compliance case. The software does not report allowable stresses.

SUS

Sustained case.

EXP

Expansion case.

OCC

Occasional case.

FAT

Fatigue case.

NOTE You must also specify the number of Load Cycles for load cases with a FAT stress type.

HGR

Spring hanger design case. These are load cases that CAESAR II uses internally to design and select spring hangers. Results are not available for these cases.

HYD

Hydro test case. Select hanger status. For a hydrotest case, the default hanger status is rigid or locked.

CRP

Creep Loading case. Code standards such as EN-13480 define a creep stress range for operating conditions, which is defined by the life of the material. In CAESAR II, CRP is a scalar combination of one SUS case and one EXP case. The software sets Output Type to **Stress**.

The software calculates CRP stresses according to EN-13480. If you specify additional load multipliers, the software applies these as additional scale factors. Other codes also use the EN-13480 method.

NOTES

- You must manually add a CRP case.
- The default combination method is **Scalar**. You can also set Combination Method to **MAX**.

K1P

KHK Level 1 (seismic code) primary longitudinal stress for the HPGSL and JPI piping codes. The longitudinal stress is due to pressure, weight, and design seismic force.

NOTE The software treats this stress type as **OCC** for other piping codes.

K1SR

KHK Level 1 (seismic code) secondary cyclic stress range for the HPGSL and JPI piping codes. The cyclic stress range is due to design force and support movement.

NOTE The software treats this stress type as **EXP** for other piping codes.

K2P

KHK Level 2 (seismic code) primary longitudinal stress for the HPGSL and JPI piping codes. The longitudinal stress is due to internal pressure, weight, seismic force, and response displacement.

NOTE The software treats this stress type as **OCC** for other piping codes.

K2SA

KHK Level 2 (seismic code) secondary cyclic stress amplitude for the HPGSL and JPI piping codes. The cyclic stress amplitude is due to seismic force and response displacement.

NOTE The software treats this stress type as **EXP** for other piping codes.

K2SR

KHK Level 2 (seismic code) secondary cyclic stress range for the HPGSL and JPI piping codes. The cyclic stress range is due to seismic force and response displacement.

NOTE The software treats this stress type as **EXP** for other piping codes.

K2L

KHK Level 2 (seismic code) liquefaction for the HPGSL and JPI piping codes. Liquefaction is the angular displacement corresponding to a maximum equivalent plastic strain of 5% (in degrees).

NOTE The software treats this stress type as **EXP** for other piping codes.

LMST

For DNV 2017, only use the **LMST** limit state for combined load cases where local buckling for the combined loading criteria is calculated. For more information, see section 5.4.6 of DNVGL-ST-F101.

MDMT

For B31.3, the minimum design metal temperature.

Standard Reports

There are a variety of report options that can be selected for review.

Bends KHK2 Report

Reports bend behavior due to KHK Level 2 loading conditions. The software generates a separate report for each selected load case. When theta (θ) bend values exceed the allowable theta, the check fails, and the failing bends display in red.

CAESAR II
Job Name:
Licensed To:
BENDS KHK2 EVALUATION REPORT: Bends on Elements
CASE 5 (K2L) WW+P1

Bend Node	h	Bend Angle deg.	Delta ThetaX deg.	Delta ThetaY deg.	Delta ThetaZ deg.	Theta deg.	Allowable Theta deg.	Ratio %
Bends KHK2 Evaluation CHECK FAILED : LOADCASE 5 (K2L) WW+P1								
50	0.0826	90.0000	-19.1987	0.0000	9.0872	21.2407	7.6543	277.4992
80	0.2243	90.0000	-11.4928	-5.0816	9.9664	16.0386	4.8328	331.8724
410	0.1168	90.0000	-0.0217	0.0000	0.0000	0.0217	6.5240	0.3324
300	0.1168	45.0000	-0.7286	0.0000	1.3818	1.5621	3.2620	47.8892
310	0.1168	45.0000	-0.4087	-0.0000	0.1452	0.4337	3.2620	13.2951
200	0.1052	45.0000	-0.0000	0.0000	0.3923	0.3923	3.4234	11.4586
210	0.0526	45.0000	-0.0000	0.0000	0.5402	0.5402	4.7091	11.4724
100	0.1052	90.0000	0.0718	0.0000	0.7816	0.7849	6.8468	11.4641
110	0.1578	90.0000	-0.6266	-0.0560	0.5123	0.8113	5.6818	14.2788
120	0.1578	90.0000	-0.0986	0.0535	0.2011	0.2303	5.6818	4.0527

NOTES

- The software only includes bends in this report.

- Use the **Filters** feature to sort reports containing fields from more than one class. For more information, see Filter Reports.

Bends KHK2 Summary Report

Reports a summary of bend behavior for multiple load cases due to KHK Level 2 loading conditions. When theta (θ) bend values exceed the allowable theta, the check fails, and the failing bends display in red.

```
CAESAR II
Job Name: [illegible]
Licensed To: [illegible]
BEND KHK2 SUMMARY REPORT: KHK Level 2 Bend Evaluation
Various Load Cases

 Bend   Load Case  K In-Plane  K In-Plane  K Out-Plane  K Average   Theta    Allowable Theta  Ratio   Status
 Node               Opening     Closing                              deg.         deg.          %

LOAD CASE DEFINITION KEY

CASE 3 (K2L) W+D1+T1+P1+H
CASE 4 (K2L) W+P1+H

KHK Level 2 Legend: P  = Bend is Plastic; bend deflection exceeds elastic limit


 Bends KHK2 Evaluation CHECK PASSED     : LOADCASE 3 (K2L) W+D1+T1+P1+H
 Bends KHK2 Evaluation CHECK PASSED     : LOADCASE 4 (K2L) W+P1+H

   30
        3(K2L)    8.178      8.178      8.178      8.178      0.120      4.834      2.5
        4(K2L)    8.178      8.178      8.178      8.178      0.016      4.834      0.3
          MAX     8.178/L3   8.178/L3   8.178/L3   8.178/L3   0.120/L3   4.834/L3   2.5/L3

   35
        3(K2L)    8.178      8.178      8.178      8.178      0.137      4.834      2.8
        4(K2L)    8.178      8.178      8.178      8.178      0.025      4.834      0.5
          MAX     8.178/L3   8.178/L3   8.178/L3   8.178/L3   0.137/L3   4.834/L3   2.8/L3

  135
        3(K2L)    8.178      8.178      8.178      8.178      0.098      4.834      2.0
        4(K2L)    8.178      8.178      8.178      8.178      0.017      4.834      0.4
          MAX     8.178/L3   8.178/L3   8.178/L3   8.178/L3   0.098/L3   4.834/L3   2.0/L3
```

NOTES

- The software only includes bends in this report.
- Use the **Filters** feature to sort reports containing fields from more than one class. For more information, see Filter Reports.

Code Compliance Report

Reports stress checks for multiple load cases. The report shows the stress calculation for all selected load cases on an element-by-element basis.

CAESAR II
Job Name:
Licensed To:
CODE COMPLIANCE REPORT: Code Stresses on Elements
Various Load Cases

Load Case	From Node	Code Stress lb./sq.in.	Allowable Stress lb./sq.in.	Ratio %	To Node	Code Stress lb./sq.in.	Allowable Stress lb./sq.in.	Ratio %	Piping Code

LOAD CASE DEFINITION KEY

CASE 3 (OPE) W+D1+T1+P1+H
CASE 4 (SUS) W+P1+H
CASE 5 (EXP) L5=L3-L4

Note: This report indicates mechanical stresses. For component stresses per piping code see the Stress Components reports.
Piping Code: B31.3 = B31.3 -2016, Jan 31, 2017

*** CODE COMPLIANCE EVALUATION PASSED ***

Highest Stresses: (lb./sq.in.)

Ratio (%):	12.0	@Node 33	LOADCASE: 5 (EXP) L5=L3-L4
Code Stress:	5434.7	Allowable Stress:	45417.4
Axial Stress:	464.3	@Node 28	LOADCASE: 4 (SUS) W+P1+H
Bending Stress:	5384.3	@Node 33	LOADCASE: 5 (EXP) L5=L3-L4
Torsion Stress:	951.1	@Node 134	LOADCASE: 5 (EXP) L5=L3-L4
Hoop Stress:	415.0	@Node 10	LOADCASE: 4 (SUS) W+P1+H
Max Stress Intensity:	6143.9	@Node 34	LOADCASE: 3 (OPE) W+D1+T1+P1+H

Load Case	From Node	Code Stress lb./sq.in.	Allowable Stress lb./sq.in.	Ratio %	To Node	Code Stress lb./sq.in.	Allowable Stress lb./sq.in.	Ratio %	Piping Code
3(OPE)	5	2655.0	0.0	0.0	10	4872.1	0.0	0.0	B31.3
4(SUS)		1113.7	17300.0	6.4		1577.8	17300.0	9.1	B31.3
5(EXP)		2348.8	45511.3	5.2		4248.8	45047.2	9.4	B31.3
3(OPE)	10	4210.8	0.0	0.0	15	2162.9	0.0	0.0	B31.3
4(SUS)		1085.3	17300.0	6.3		809.2	17300.0	4.7	B31.3
5(EXP)		3862.6	45539.7	8.5		1854.9	45815.8	4.0	B31.3
3(OPE)	15	0.0	0.0	0.0	20	0.0	0.0	0.0	B31.3
4(SUS)		0.0	0.0	0.0		0.0	0.0	0.0	B31.3
5(EXP)		0.0	0.0	0.0		0.0	0.0	0.0	B31.3

Cumulative Usage Report

Reports the combined impact of simulating selected fatigue loadings for the selected load cases.

NOTE The **Cumulative Usage** report is available only when there are one or more fatigue-type load cases present.

```
 CAESAR II  CUMULATIVE USAGE    FILE:FATIGUE

                                               From  Allow. Usage    To  Allow. Usage
Load Case                               Cycles Node  Cycles Ratio   Node Cycles Ratio
------------------------------------------------------------------------------------------
          ****       CUMULATIVE USAGE EVALUATION PASSED

          HIGHEST USAGE RATIO IS    0.87 AT NODE HINODE
          MINIMUM ALLOWABLE CYCLES IS    12338


CASE 10 (FAT) L10=L6-L4                    0      5        0   0.00  start        0   0.00
CASE 11 (FAT) L11=L6-L7                    0      5        0   0.00  start        0   0.00
CASE 12 (FAT) L12=L6-L8                    0      5        0   0.00  start        0   0.00
CASE 13 (FAT) L13=L3-L8                    0      5        0   0.00  start        0   0.00
  TOTAL:                                          5            0.00  start            0.00

CASE 10 (FAT) L10=L6-L4                  200 start11201984     0.00     1511200832   0.00
CASE 11 (FAT) L11=L6-L7                11800 start11220608     0.00     1511219904   0.00
CASE 12 (FAT) L12=L6-L8               188000 start INFINTY     0.00     15 INFINTY   0.00
CASE 13 (FAT) L13=L3-L8                12000 start INFINTY     0.00     15 INFINTY   0.00
  TOTAL:                                     start             0.00     15           0.00

CASE 10 (FAT) L10=L6-L4                  200    1510302976     0.00     2010303424   0.00
```

Displacements Report

Reports translations and rotations for each degree of freedom at each node in the model.

CAESAR II
Job: [redacted]
Licensed To: [redacted]
DISPLACEMENTS REPORT: Nodal Movements
CASE 1 (OPE) W+T1+P1

NODE	DX in.	DY in.	DZ in.	RX deg.	RY deg.	RZ deg.
10	0.0000	-0.0000	0.0000	-0.0000	-0.0000	-0.0000
20	0.0000	-0.0000	-0.0185	-0.0005	-0.0006	-0.0005
30	0.0263	-0.0485	-0.1879	-0.1210	-0.0025	-0.1607
38	-0.0188	-0.0543	-0.6886	0.3347	-0.1346	0.0662
39	-0.0060	-0.0137	-0.6920	0.5650	-0.2546	0.0438
40	0.0000	0.0244	-0.6492	0.7678	-0.3064	0.0323
48	-0.0324	0.2193	-0.0315	0.6581	-0.6714	0.0350
49	-0.0265	0.2384	0.0286	0.4139	-0.6483	0.0109
50	-0.0081	0.2459	0.0879	0.3067	-0.6613	-0.0112
58	0.0634	0.2415	0.2823	-0.0061	-0.6640	-0.0146
59	0.0817	0.2331	0.3330	-0.1153	-0.6578	-0.0020
60	0.0899	0.2148	0.3713	-0.3614	-0.6840	0.0190
63	0.1077	0.0955	0.5999	-0.5315	-0.4724	0.0621

NOTE Use the **Filters** feature to sort reports containing fields from more than one class. For more information, see Filter Reports.

DNV Components Report

Displays the minimum thickness and utility value for Burst, Collapse, and each LCC and DCC load combination.

Each node with a utility value greater than 1.0 displays in red.

```
CAESAR II
Job Name:
Licensed To:
DNV COMPONENTS REPORT: DNV Components on Elements
CASE 7 (LMST) L7=L2+L5
```

Node	Input Wall Thickness in.	Tmin 1 in.	Utility 1	Tmin 2 in.	Utility 2	Tmin 3 in.	Utility 3	Tmin 4 in.	Utility 4	Warnin

```
Note: The DNV Components report shows results for the DNV 2017 edition and beyond. For additional details on DNV piping, see the DNV Details report.

Warning:
(1) Invalid condition: Internal over pressure.
(2) Minimum wall thickness less than Outside Diameter/2 was not found.
(3) Input diameter-to-wall-thickness ratio is outside range 15<=D/t2<=45 (eq. 5.16 & 5.19)
(4) Ratio Sd/Sp>0.4 (SSd = design effective axial force load effect,
    Sp = plastic capability of pipe, eq. 5.19)

Utility 1, Tmin 1: Load combination LCC-a (eq. 5.19, 5.28)
Utility 2, Tmin 2: Load combination LCC-b (eq. 5.19, 5.28)
Utility 3, Tmin 3: Load combination DCC-a (eq. 5.19, 5.28)
Utility 4, Tmin 4: Load combination DCC-b (eq. 5.19, 5.28)

 DNV Evaluation CHECK FAILED      : LOADCASE 7 (LMST) L7=L2+L5

Highest DNV Value: (in.)
Utility 1       34.251    Tmin 1              N/A    @Node     10
Utility 2       28.781    Tmin 2              N/A    @Node     10
Utility 3        0.002    Tmin 3            0.000    @Node     10
Utility 4        0.002    Tmin 4            0.000    @Node     10
```

Node	Input Wall Thickness in.	Tmin 1 in.	Utility 1	Tmin 2 in.	Utility 2	Tmin 3 in.	Utility 3	Tmin 4 in.	Utility 4	Warning
10	0.500		34.251		28.781	0.000	0.002	0.000	0.002	2
18	0.500	0.131	0.035	0.124	0.030	0.000	0.002	0.000	0.002	
20	0.500	0.121	0.027	0.115	0.023	0.000	0.002	0.000	0.002	
30	0.500	0.056	0.000	0.056	0.000	0.000	0.002	0.000	0.002	

NOTE Use the **Filters** feature to sort reports containing fields from more than one class. For more information, see Filter Reports.

DNV Details Report

Displays forces, moments, and strains for DNV design criteria.

Nodes with values that do not meet DNV criteria display in red.

CAESAR II
Job Name: [illegible]
Licensed To: [illegible]
DNV DETAILS REPORT: DNV Details on Elements
CASE 6 (LMST) L6=L1+L5

Note: The DNV Details report shows results for DNV 2017 Edition and beyond.

P int 1: Internal Pressure for e.g. Burst Operation, Collapse, LCC-a, DCC-a
P int 2: Internal Pressure for Burst System Test, Propagation buckling, LCC-b, DCC-b

Node	Depth ft.	P ext lb./sq.in.	P int 1 lb./sq.in.	P int 2 lb./sq.in.	Functional Axial Force lb.	Environ. Axial Force lb.	Functional Moment ft.lb.	Environ. Moment ft.lb.	Functional Strain %	Environ. Strain %
10	400.000	177.589	0.000	0.000	0	0	946759.9	10.2	0.000	0.000
18	400.000	177.589	0.000	0.000	0	0	30428.3	0.3	0.000	0.000
20	399.920	177.553	0.000	0.000	2459	0	26619.4	0.3	0.000	-0.000
30	395.000	175.369	0.000	0.000	0	0	0.0	0.0	0.000	-0.000

NOTE Use the **Filters** feature to sort reports containing fields from more than one class. For more information, see Filter Reports.

DNV Thickness Overview Report

Displays the wall thicknesses for DNV design criteria.

Nodes with wall thicknesses that do not meet a DNV criteria display in red.

CAESAR II
Job Name: [illegible]
Licensed To: Licensed To: [illegible]
DNV THICKNESS OVERVIEW REPORT: DNV Thickness Overview on Elements
Various Load Cases

LOAD CASE DEFINITION KEY

CASE 1 (OCC) WNC+T1
CASE 2 (OCC) WW+T1
CASE 3 (OPE) WW+T1+P2
CASE 4 (OCC) W+T1+P1
CASE 5 (OCC) 0.00001W+WAV1
CASE 6 (LMST) L6=L1+L5
CASE 7 (LMST) L7=L2+L5
CASE 8 (LMST) L8=L3+L5
CASE 9 (LMST) L9=L4+L5

Warning - Minimum wall thickness less than Outside Diameter/2 was not found.

Node	Load Case	Input Wall Thickness in.	Max of All Tmin in.	Burst Operation Tmin in.	Burst System Test Tmin in.	Collapse Tmin in.	Propagation Buckling Tmin in.	LCC-a Tmin in.	LCC-b Tmin in.	DCC-a Tmin in.	DCC-b Tmin in.
10	MAX	0.500	Warning/L6	0.349/L4	0.320/L3	0.209/L1	0.396/L1	Warning/L6	Warning/L6	0.000/L6	0.000/L6
18	MAX	0.500	0.396/L1	0.349/L4	0.320/L3	0.209/L1	0.396/L1	0.265/L8	0.263/L8	0.000/L6	0.000/L6
20	MAX	0.500	0.396/L1	0.349/L4	0.320/L3	0.209/L1	0.396/L1	0.263/L8	0.261/L8	0.000/L6	0.000/L6
30	MAX	0.500	0.394/L1	0.350/L4	0.320/L3	0.208/L1	0.394/L1	0.253/L8	0.253/L8	0.000/L6	0.000/L6

NOTE Use the **Filters** feature to sort reports containing fields from more than one class. For more information, see Filter Reports.

Flange NC-3658.3 Report

Displays flange data after completing an in-line flange evaluation analysis using the ASME NC-3658.3 Calculation Method for B16.5 Flanged Joints with High Strength Bolting.

The report displays input items along with the calculated moments and stresses or equivalent pressure for each node where the flange evaluation was requested. This is an elemental type report, and flanges may be defined on either end of the element. As a result, some lines in the report with no corresponding output are blank.

```
CAESAR II
Job: [illegible]
Licensed To: [illegible]
FLANGE NC-3658.3 REPORT: Flange (NC-3658.3 Method)
CASE 3 (OPE) W+T1+P1+H
```

NODE	Torsion Moment ft.lb.	Bending Moment ft.lb.	G/C in.	AB sq.in.	Flange Stress lb./sq.in.	Allowable Pressure /Stress	Ratio %
50	0.0	3608.1	18.75	5.0280	440.9	30023.5	1.47
55							
55							
60	0.0	2562.1	18.75	5.0280	313.1	30023.5	1.04

Flange Peq Report

Displays flange data after completing an in-line flange evaluation analysis using the Kellogg Equivalent Pressure Method (P_{eq}).

The report displays input items along with the calculated moments and stresses or equivalent pressure for each node where the flange evaluation was requested. This is an elemental type report, and flanges may be defined on either end of the element. As a result, some lines in the report with no corresponding output are blank.

```
CAESAR II
Job: [illegible]
Licensed To: [illegible]
FLANGE PEQ REPORT: Flange (Equiv Pressure Method)
CASE 3 (OPE) W+T1+P1+H
```

NODE	Axial Force lb.	Bending Moment ft.lb.	G/C in.	P Equivalent lb./sq.in.	Rating Temperature F	Allowable Pressure /Stress	Ratio %
1	4537	46247.2	15.75	98.51	70.00	913.75	10.78
5	3202	46247.2	15.75	97.94	70.00	913.75	10.72
90	0	3608.1	15.75	40.97	70.00	1800.00	2.28
95							
95							
100	1241	0.0	15.75	36.79	70.00	1800.00	2.04

NOTE The software uses the operating pressure (in the selected flange analysis load case) as the design pressure. For the example report shown here, the P_{eq} result is generated using the pressure P1 from the selected operating load case 3.

Global Element Forces Extended Report

Reports forces and moments, including shear forces, bending moments, and torsional moments, on the piping for each node in the model. The report also displays element names.

```
CAESAR II
Job Name:
Licensed To:
GLOBAL ELEMENT FORCES EXTENDED REPORT: Forces on Elements
CASE 3 (OPE) W+T1+P1+H
```

Node	Shear Force lb.	Bending Moment ft.lb.	Torsion Moment ft.lb.	FX lb.	FY lb.	FZ lb.	MX ft.lb.	MY ft.lb.	MZ ft.lb.	
5	0	43603.2	0.0	-0	3043	0	41947.2	0.0	-11902.7	
10	2762	37710.8	-11902.7	0	-2762	-0	-37710.8	-0.0	11902.7	
10	2762	37710.8	11902.7	-0	2762	0	37710.8	0.0	-11902.7	
15	1713	9239.2	7094.9	0	-1713	-0	-7094.9	-0.0	9239.2	Horiz 1
15	1713	9239.2	-7094.9	-0	1713	0	7094.9	0.0	-9239.2	
20	0	8516.4	-0.0	0	-884	-0	-7094.9	-0.0	-4710.7	Horiz 2
20	0	8516.4	0.0	-0	2559	0	7094.9	0.0	4710.7	
22	0	8516.4	-0.0	0	-1573	-0	-7094.9	-0.0	-4710.7	Vert 1
22	0	8516.4	0.0	-0	3348	0	7094.9	0.0	4710.7	
25	1970	7760.2	7094.9	0	-1970	-0	-7094.9	-0.0	-7760.2	Vert 2

NOTE Use the **Filters** feature to sort reports containing fields from more than one class. For more information, see Filter Reports.

Global Element Forces Report

Reports forces and moments on the piping for each node in the model. The report also displays element names.

CAESAR II
Job Name: [redacted]
Licensed To: [redacted]
GLOBAL ELEMENT FORCES REPORT: Forces on Elements
CASE 3 (OPE) W+T1+P1+H

Node	FX lb.	FY lb.	FZ lb.	MX ft.lb.	MY ft.lb.	MZ ft.lb.	
5	-0	3043	0	41947.2	0.0	-11902.7	
10	0	-2762	-0	-37710.8	-0.0	11902.7	
10	-0	2762	0	37710.8	0.0	-11902.7	
15	0	-1713	-0	-7094.9	-0.0	9239.2	Horiz 1
15	-0	1713	0	7094.9	0.0	-9239.2	
20	0	-884	-0	-7094.9	-0.0	-4710.7	Horiz 2
20	-0	2559	0	7094.9	0.0	4710.7	
22	0	-1573	-0	-7094.9	-0.0	-4710.7	Vert 1
22	-0	3348	0	7094.9	0.0	4710.7	
25	0	-1970	-0	-7094.9	-0.0	-7760.2	Vert 2

NOTE Use the **Filters** feature to sort reports containing fields from more than one class. For more information, see Filter Reports.

Local Element Forces Report

Reports forces and moments translated into the CAESAR II local coordinate system. The report also displays element names. For more information, see Local Coordinates in the *Technical Discussions* section of the *CAESAR II User's Guide*.

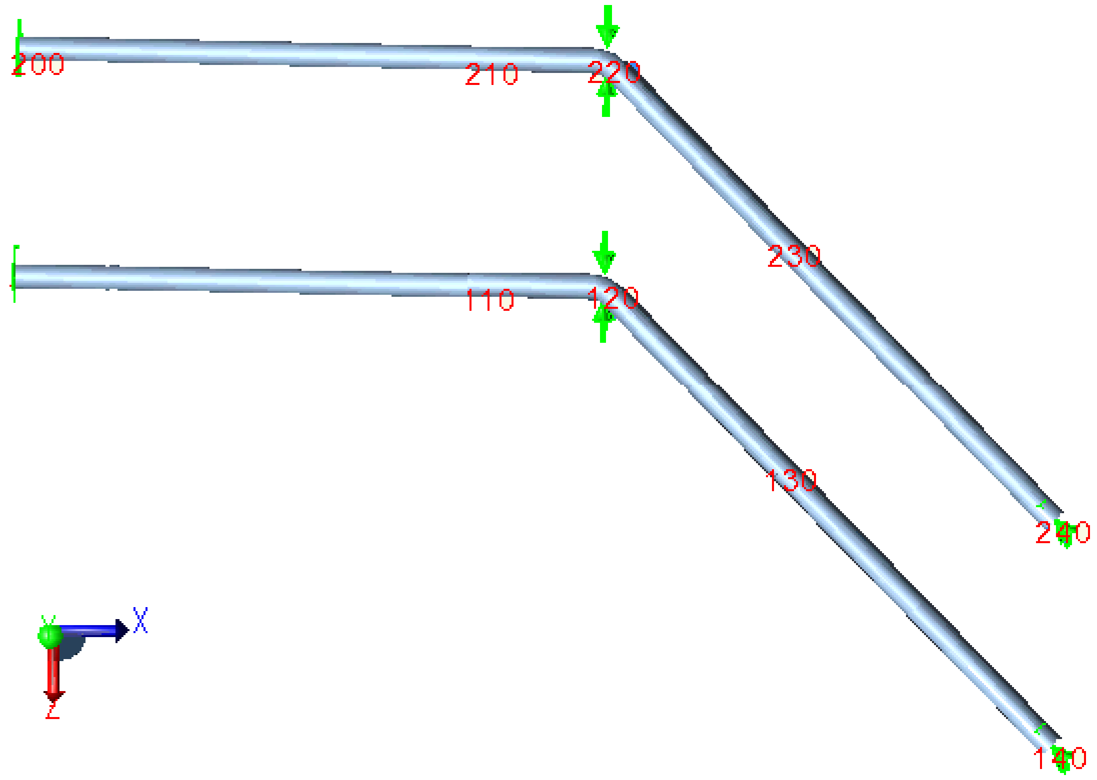
CAESAR II
Job Name:
Licensed To:
LOCAL ELEMENT FORCES REPORT: Forces on Elements
CASE 3 (OPE) W+D1+T1+P1+H

Node	fx lb.	fy lb.	fz lb.	mx ft.lb.	my ft.lb.	mz ft.lb.	Element Name
5	206	-136	-235	1539.8	2755.3	1519.8	
10	-97	235	-136	-1539.8	-1792.2	2284.4	
10	-378	-189	80	1414.2	1089.4	-2125.0	
15	410	80	189	-1414.2	-2014.8	-1135.8	
15	-410	-80	-189	1414.2	2014.8	1135.8	
20	956	80	189	-1414.2	-1577.6	-1319.9	
20	-956	-80	-189	1414.2	1577.6	1319.9	
25	988	-189	80	-1414.2	1366.3	-1467.3	
25	-965	235	-136	1539.8	-1508.8	1464.9	
28	1465	136	235	-1539.8	693.3	-2757.6	
28	737	-235	136	1539.8	2757.6	693.3	
29	-587	235	394	-1647.9	-2646.5	432.1	
29	587	-235	-394	1647.9	2646.5	-432.1	
30	-136	235	651	-928.7	-2211.7	1304.3	

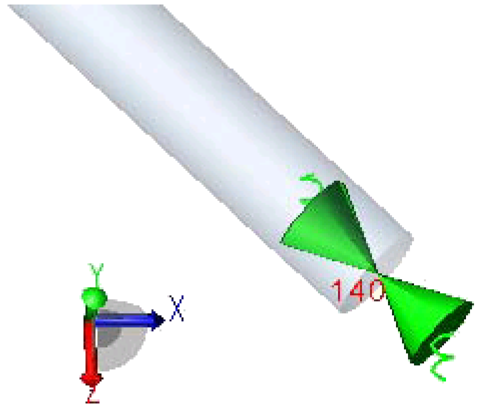
NOTE Use the **Filters** feature to sort reports containing fields from more than one class. For more information, see Filter Reports.

Local Restraints Report

Generates a restraint report where the loads and moments are aligned with the local element coordinate system. This is particularly useful when addressing skewed nozzles, where the axial, longitudinal and circumferential results are needed. As an example, consider the small system shown below:



This system consists of two small horizontal lines anchored at both ends. The last element of each line is skewed 45 degrees in the X-Z plane. At the end of this skewed line is an axial restraint, as illustrated in the following figure:



The typical **Global Restraint** report for this system displays in the following table. At node 140, this report shows two equal loads in the (global) X and Z directions. These values (24,463) are the global component loads acting on the skewed restraint. The actual magnitude of the restraint load, acting in-line with the pipe can be found by performing the SRSS of these component loads, which yields 34595. This value is the load on the restraint acting axially with the pipe.

Operating Case Restraint Loads – Global Coordinate System

NODE	FX lb.	FY lb.	FZ lb.	MX ft.lb.	MY ft.lb.	MZ ft.lb.	
100	-24463	**-514**	66	1340.5	-273.3	-6418.6	Rigid ANC
119	0	0	-24528	0.0	0.0	0.0	Rigid Z
140	**24463**	0	**24463**	0.0	0.0	0.0	Flex X
200	-24463	-514	66	1340.5	-273.3	-6418.6	Rigid ANC
219	0	0	-24528	0.0	0.0	0.0	Rigid Z
240	24463	0	24463	0.0	0.0	0.0	Flex X

The process of performing SRSS or sine/cosine operations to obtain restraint loads in the element coordinate system can be tedious. As an alternative, generate a restraint report where all loads are aligned with the associated element coordinate system. The report for the same small job displays in the table below.

Operating Case Restraint Loads – Local Element Coordinate System

NODE	fx lb.	fy lb.	fz lb.	mx ft.lb.	my ft.lb.	mz ft.lb.	
100	-24463	66	**514**	1340.5	-6418.6	273.3	Rigid ANC
119	0	-24528	0	0.0	0.0	0.0	Rigid Z
140	**34595**	0	0	0.0	0.0	0.0	Flex X
200	-24463	66	514	1340.5	-6418.6	273.3	Rigid ANC
219	-17344	-17344	0	0.0	0.0	0.0	Rigid Z
240	34595	0	0	0.0	0.0	0.0	Flex X

In reviewing the relationship between the local versus global restraint loads note the following:

The global FY (vertical) load at node 100 of -514 translates to a local fz load. For details on the global to local coordinate system relations, see Technical Discussions. (These two values are shown in the tables in **bold**.)

At node 140, the skewed axial restraint, the first table showing the global coordinate system loads reports the two equal component loads. The second table showing the local loads, reports only the resultant axial load at the restraint. (These values are shown in the tables in **bold**.)

MDMT Report

Reports the applicable stress ratio and the amount of further temperature reduction allowed by the B31.3 code without requiring impact testing.

CAESAR II
Job Name: [illegible]
Licensed To: [illegible]
MDMT REPORT: Reduced Impact Test Exemption Temperature on Elements without Impact Testing
CASE 2 (MDMT) W+T1+P1+F1

Note: Calculations are performed according to B31.3 Section 323.2.2 Para (a), (b 1), (b 3), (d), (e), (g), and (h).
Note: Only pipes, bends, reducers, and tees are reported.
Note: You should also consider other factors, such as: shock loading, thermal bowing, dissimilar welds, elevated temperatures, and welding procedures.

Warnings:
3: Final MDMT cannot be set below the reduction limit. Status is Impact.
4: Base MDMT is already below the reduction limit. Status is Impact.
5: Stress Ratio is >= 1.0 or allowable stress is zero. Status is Impact.
-2: Pipe Thk is out of range. Status is Unknown.
-3: MDMT is not available for this material. Status is Unknown.
-4: Thermal expansion coefficient is defined in the temperature field. Status is Unknown.

Node	Pipe Thk in.	Mat ID	Curve	Piping Code	Min [Ta,T1~T9] F	Base MDMT F	Stress Ratio	Reduction F	Final MDMT F	Status	Warnings
10	0.680	110	A	B31.3	-30.0	47.4	0.86	14.1	33.3	Impact	
18	0.680	110	A	B31.3	-30.0	47.4	0.36	136.4	-55.0	OK	
18	0.680	110	A	B31.3	-30.0	47.4	0.36	136.4	-55.0	OK	
19	0.680	110	A	B31.3	-30.0	47.4	0.48	67.8	-20.4	Impact	
19	0.680	110	A	B31.3	-30.0	47.4	0.48	67.8	-20.4	Impact	
20	0.680	110	A	B31.3	-30.0	47.4	0.49	66.2	-18.8	Impact	
20	0.680	372	D	B31.3	-110.0	-43.5	0.26	217.0	-155.0	OK	
30	0.680	372	D	B31.3	-110.0	-43.5	0.37	132.9	-55.0	Impact	3
30	0.680	372	D	B31.3	-110.0	-43.5	0.34	156.5	-55.0	Impact	3
40	0.400	372	D	B31.3	-110.0	-55.0	0.53	53.3	-55.0	Impact	4

Nozzle Check Report

Defines the appropriate force/moment limits on a specified nozzle.

CAESAR II
Job: [illegible]
Licensed To: [illegible]
NOZZLE CHECK REPORT: Nozzle Loads Screening
Various Load Cases

NODE		fa lb.	fb lb.	fc lb.	Resultant Force lb.	ma ft.lb.	mb ft.lb.	mc ft.lb.	Resultant Moment ft.lb.	

LOAD CASE DEFINITION KEY

CASE 1 (OPE) W+T1+P1
CASE 2 (SUS) W+P1

NODE		fa lb.	fb lb.	fc lb.	Resultant Force lb.	ma ft.lb.	mb ft.lb.	mc ft.lb.	Resultant Moment ft.lb.	
10	Absolute Method									
	Limits	1500	1000	1200	0	3700.0	2800.0	1800.0	0.0	
	1(OPE)	-1065	-278	494	0	197.9	-3173.1	-409.7	0.0	*
	2(SUS)	0	-293	-0	0	-332.2	2.8	-468.6	0.0	
100	SRSS Method									
	Limits	850	1100	700	1556	1700.0	1300.0	900.0	2321.6	
	1(OPE)	6478	-14	80	6479	161.5	42.0	127.7	210.1	*
	2(SUS)	34	0	-1	34	0.3	-0.2	-0.4	0.6	
200	Unity Check Method									
	Limits	500	800	600		700.0	400.0	1000.0		
	1(OPE)	1079	-413	94	2.832	89.6	66.2	1400.1	1.694	*
	2(SUS)	-0	-0	164	0.274	-36.6	496.9	1.1	1.296	*

Data for each load case reported is a result of calculation (and can also be viewed on a Local Restraints report). The **Limits** shown in the report are the values from the input. Similarly, the **Comparison** method also reflects the input setting. The loads shown are the loads on the nozzle for the indicated load cases. If any load exceeds its corresponding allowable load, then the entire line is shown in red (with an asterisk at the far right in the event the report is printed in black and white.)

The **Resultant** column reports the resultant forces and moments for the **SRSS Comparison** method, and the unity check value for the **Unity Check** method.

Restraint Summary Extended Report

Provides force, moment, and displacement data for all valid selected load cases together on one report.

```
CAESAR II
Job Name: [illegible]
Licensed To: [illegible]
RESTRAINT SUMMARY EXTENDED REPORT: Loads On Restraints
Various Load Cases

 Node   Load Case        FX          FY          FZ          MX          MY          MZ          DX          DY          DZ
                         lb.         lb.         lb.       ft.lb.      ft.lb.      ft.lb.        in.         in.         in.

LOAD CASE DEFINITION KEY

CASE 3 (OPE) W+D1+T1+P1+H
CASE 4 (SUS) W+P1+H
CASE 5 (EXP) L5=L3-L4


   5           TYPE=Displ. Reaction;
        3(OPE)        136        -206        -235     -2755.3     -1539.8      1519.8      0.0000      0.0770      0.0460
        4(SUS)          8         337          -3        15.3        30.3      1110.8      0.0000      0.0000     -0.0000
        5(EXP)        129        -544        -232     -2770.6     -1570.1       408.9      0.0000      0.0770      0.0460
          MAX      136/L3     -544/L5     -235/L3  -2770.6/L5  -1570.1/L5   1519.8/L3   0.0000/L3   0.0770/L3   0.0460/L3

  28           TYPE=Prog Design  VSH;
        3(OPE)          0       -2202           0         0.0         0.0         0.0     -0.1148      0.7510     -0.0560
        4(SUS)          0       -2540           0         0.0         0.0         0.0     -0.0455      0.0009      0.0021
        5(EXP)          0         338           0         0.0         0.0         0.0     -0.0693      0.7501     -0.0581
          MAX                -2540/L4                                                  -0.1148/L3   0.7510/L3  -0.0581/L5
```

NOTE Use the **Filters** feature to sort reports containing fields from more than one class. For more information, see Filter Reports.

Restraint Summary Report

Reports the sum of force and moment reactions for all selected load cases and for all restraints at a single node.

```
CAESAR II
Job: [illegible]
Licensed To: [illegible]
RESTRAINT SUMMARY REPORT: Loads On Restraints
Various Load Cases

  NODE   Load Case       FX           FY           FZ           MX           MY           MZ
                         lb.          lb.          lb.        ft.lb.       ft.lb.       ft.lb.

LOAD CASE DEFINITION KEY

CASE 1 (OPE) W+T1+P1
CASE 2 (SUS) W+P1
CASE 3 (EXP) L3=L1-L2


  10           Rigid ANC
         1(OPE)       986         -336         1730      -1524.8      -1969.5       -982.8
         2(SUS)        -1         -148          -36       -268.7         22.0       -101.4
         3(EXP)       987         -188         1765      -1256.1      -1991.5       -881.4
           MAX     987/L3      -336/L1      1765/L3   -1524.8/L1   -1991.5/L3    -982.8/L1
```

NOTE Use the **Filters** feature to sort reports containing fields from more than one class. For more information, see Filter Reports.

Restraints Extended Report

Reports forces and moments on each restraint in the model, including resultant forces and moments. The software generates a separate report for each selected load case.

CAESAR II
Job Name: [illegible]
Licensed To: [illegible]
RESTRAINTS EXTENDED REPORT: Loads On Restraints
CASE 3 (OPE) W+D1+T1+P1+H

Node	FX lb.	FY lb.	FZ lb.	Resultant Force lb.	MX ft.lb.	MY ft.lb.	MZ ft.lb.	Resultant Moment ft.lb.	Restraint Type/Tag
5	136	-206	-235	342	-2755.3	-1539.8	1519.8	3503.2	TYPE=Displ. Reaction;
28	0	-2202	0	2202	0.0	0.0	0.0	0.0	TYPE=Prog Design VSH;
33	0	-948	0	948	0.0	0.0	0.0	0.0	TYPE=Rigid +Y;
40	0	-721	0	721	0.0	0.0	0.0	0.0	TYPE=Rigid Y;
40	0	0	0	0	0.0	1166.8	0.0	1166.8	TYPE=Flex RY;
40	-136	0	0	136	0.0	0.0	0.0	0.0	TYPE=Rigid X;
40	0	0	0	0	-2473.2	0.0	0.0	2473.2	TYPE=Flex RX;
40	0	0	235	235	0.0	0.0	0.0	0.0	TYPE=Flex Z;
40	0	0	0	0	0.0	0.0	1418.1	1418.1	TYPE=Rigid RZ;
140	0	-68	0	68	0.0	0.0	0.0	0.0	TYPE=Rigid +Y;
6000	-136	-721	235	770	-2473.2	1166.8	1418.1	3080.4	TYPE=Displ. Reaction;

NOTE Use the **Filters** feature to sort reports containing fields from more than one class. For more information, see Filter Reports.

Restraints Report

Reports the force and moment reactions for each restraint at a node. The software generates a separate report for each selected load case.

CAESAR II
Job: [illegible]
Licensed To: [illegible]
RESTRAINTS REPORT: Loads On Restraints
CASE 1 (OPE) W+T1+P1

NODE	FX lb.	FY lb.	FZ lb.	MX ft.lb.	MY ft.lb.	MZ ft.lb.	
10	986	-336	1730	-1524.8	-1969.5	-982.8	Rigid ANC
40	0	0	0	0.0	0.0	0.0	Rigid +Y
40	139	0	0	0.0	0.0	0.0	Rigid GUI
130	0	0	-1490	0.0	0.0	0.0	Rigid +Z
190	162	-540	-115	-56.8	-225.0	1746.8	Rigid ANC
240	-1287	-358	-125	273.6	-2273.7	2476.4	Rigid ANC

NOTE Use the **Filters** feature to sort reports containing fields from more than one class. For more information, see Filter Reports.

Stress Summary Report

Reports the highest stresses and the corresponding node in a summary format for all selected load cases.

```
CAESAR II
Job Name:
Licensed To:
STRESS SUMMARY (LEGACY) REPORT: Highest Stresses Mini Statement
CASE 1 (OPE) W+T1+P1

LOAD CASE DEFINITION KEY

CASE 1 (OPE) W+T1+P1


Note: This report indicates generic stresses summary. Currently, CAESAR II does not have summary for code-defined stresses.
Piping Code: Multiple Codes
   B31.3      = B31.3 -2016, Jan 31, 2017
   B31.4      = B31.4 -2016, March 31, 2016

Legend: R - Restrained,  U - Unrestrained,  W - Inland Waterways


 NO CODE STRESS CHECK PROCESSED: LOADCASE 1 (OPE) W+T1+P1

Highest Stresses: (lb./sq.in.) LOADCASE 1 (OPE) W+T1+P1
Ratio (%):                   0.0       @Node     80
OPE Stress:             483056.1       Allowable Stress:         0.0
Axial Stress:             3584.0       @Node     20
Bending Stress:         482153.9       @Node     80
Torsion Stress:         105933.9       @Node     70
Hoop Stress:              7266.7       @Node     20
Max Stress Intensity:   486287.0       @Node     80
```

NOTE Use the **Filters** feature to sort reports containing fields from more than one class. For more information, see Filter Reports.

Stresses (Multiple Code/Allow) Report

Reports hoop stresses, longitudinal stresses, combined stresses, the code allowable, and code-stress-to-allowable ratios for piping codes with multiple failure criteria, such as transportation, FRP, and offshore codes. For more information, see Code.

NOTE This report does not display and the software generates no data for piping codes that consider only single stress criteria.

CAESAR II
Job Name: [illegible]
Licensed To: [illegible]
B31.4 STRESSES (MULTIPLE CODE/ALLOW.) REPORT: Stresses (Multiple Code/Allow.) on Elements
CASE 2 (SUS) W+P1

Node	Hoop lb./sq.in.	Hoop Allow. lb./sq.in.	Hoop Ratio	Long. lb./sq.in.	Long. Allow. lb./sq.in.	Long. Ratio	Comb. lb./sq.in.	Comb. Allow. lb./sq.in.	Comb. Ratio	Status
Piping Code: B31.4 -2016, March 31, 2016										
30	5333.3	21600.0	24.7	20328.5	27000.0	75.3	24835.0	27000.0	92.0	R
298	5333.3	21600.0	24.7	9436.9	27000.0	35.0	15688.0	27000.0	58.1	R
298	5333.3	21600.0	24.7	9436.9	27000.0	35.0	15688.0	27000.0	58.1	R
299	5333.3	21600.0	24.7	12426.0	27000.0	46.0	16104.1	27000.0	59.6	R
299	5333.3	21600.0	24.7	12426.0	27000.0	46.0	16104.1	27000.0	59.6	R
300	5333.3	21600.0	24.7	13775.0	27000.0	51.0	16074.8	27000.0	59.5	R

ISO 14692

To ensure safe design, FRP pipe (ISO 14692 2005 and ISO 14692 2017) limits axial and hoop stress pairs to those within the allowable failure envelope. ISO 14692 2017 extends the envelope to include negative-value compressive axial stress and includes stress variations related to local shell conditions (hoop bending and axial roping stress). For more information, see the code-specific note ISO-14692.

NOTE A negative ratio always means passing criteria.

Stresses Extended (Legacy) Report

Reports stress intensification factors, allowable stresses, and code stresses for each node in the model. The report displays the code stresses as a percentage when compared to the allowable stress at each node. Stresses are not computed at nodes on rigid elements or on structural steel elements. The report also displays mechanical axial stresses, bending stresses, torsional stresses, hoop stresses, and element names. The software calculates the component stresses using basic engineering principles.

NOTES

- For the **Stresses Extended (Legacy)** report, the software calculates the component stresses using basic engineering principles. To verify solutions according to piping code equations, run a Stresses Report.
- This report was called the **Stresses Extended** report in versions prior to CAESAR II 2019. Use this legacy report to benchmark your CAESAR II analyses previously run on older versions of the software.

CAESAR II
Job Name: [illegible]
Licensed To: [illegible]
STRESSES EXTENDED (LEGACY) REPORT: Stresses Extended (Legacy) on Elements
CASE 1 (OPE) W+T1+P1

Node	Axial Stress lb./sq.in.	Bending Stress lb./sq.in.	Torsion Stress lb./sq.in.	Hoop Stress lb./sq.in.	Max Stress Intensity lb./sq.in.	SIF/Index In-Plane	SIF/Index Out-Plane	Code Stress lb./sq.in.	Allowable Stress lb./sq.in.	Ratio %	Piping Code	Element Name

Note: This report indicates generic stresses. For code-defined stresses, see the Stresses report.
Piping Code: Multiple Codes
B31.3 = B31.3 -2016, Jan 31, 2017
B31.4 = B31.4 -2016, March 31, 2016

Legend: R - Restrained, U - Unrestrained, W - Inland Waterways

NO CODE STRESS CHECK PROCESSED: LOADCASE 1 (OPE) W+T1+P1

Highest Stresses: (lb./sq.in.)

Ratio (%):	0.0	@Node	80
OPE Stress:	483056.1	Allowable Stress:	0.0
Axial Stress:	3584.0	@Node	20
Bending Stress:	482153.9	@Node	80
Torsion Stress:	105933.9	@Node	70
Hoop Stress:	7266.7	@Node	20
Max Stress Intensity:	486287.0	@Node	80

Node	Axial Stress	Bending Stress	Torsion Stress	Hoop Stress	Max Stress Intensity	SIF/Index In-Plane	SIF/Index Out-Plane	Code Stress	Allowable Stress	Ratio %	Piping Code	Element Name
10	3584.0	240.0	-3389.4	7266.7	9538.8	1.000	1.000	10367.1	0.0	0.0	B31.3	
20	3584.0	11887.3	3389.4	7266.7	16891.2	1.000	1.000	17268.3	0.0	0.0	B31.3	
20	3584.0	11887.3	-4693.9	7266.7	18096.7	1.000	1.000	18731.2	0.0	0.0	B31.3	
30	3584.0	34778.5	4693.9	7266.7	39494.5	1.000	1.000	39607.3	0.0	0.0	B31.3	
30	3584.0	29459.8	-9395.0	7266.7	38012.6	1.000	1.000	38526.0	0.0	0.0	B31.3	
40	3584.0	64303.5	9395.0	7266.7	70439.9	1.000	1.000	70576.6	0.0	0.0	B31.3	

NOTE Use the **Filters** feature to sort reports containing fields from more than one class. For more information, see Filter Reports.

Stresses Report

Reports stresses at each node in the model. Also reports the highest stresses and the corresponding nodes. Stresses are not computed at nodes on rigid elements or on structural steel elements. The software calculates stress components according to the piping code. For more information, see *Code Stresses* in the *CAESAR II Quick Reference Guide*.

```
CAESAR II
Job Name: [illegible]
Licensed To: [illegible]
B31.3 STRESSES REPORT: Stresses on Elements
CASE 1 (OPE) W+T1+P1

               SLP         F/A      Bending    Torsion   SIF/Index  SIF/Index  SIF/Index  SIF/Index     Code     Allowable
  Node    lb./sq.in.  lb./sq.in. lb./sq.in. lb./sq.in.  In-Plane  Out-Plane   Torsion     Axial   lb./sq.in. lb./sq.in.   Ratio


Piping Code (1 of 2): B31.3 -2016, Jan 31, 2017

The SLP column shows the longitudinal pressure stress.


 NO Stresses Evaluation CHECK PROCESSED: LOADCASE 1 (OPE) W+T1+P1

Highest Stresses: (lb./sq.in.)
SLP                          3584.0       @Node     10
F/A                          1640.9       @Node     90
Bending                    482153.9       @Node     80
Torsion                    105933.9       @Node     70
SIF/Index In-Plane              5.0       @Node    398
SIF/Index Out-Plane             5.0       @Node    398
SIF/Index Torsion               1.0       @Node     10
SIF/Index Axial                 1.0       @Node     10


   10     3584.0        -0.0      240.0    -3389.4      1.000      1.000      1.000      1.000    10367.1
   20     3584.0        -0.0    11887.3     3389.4      1.000      1.000      1.000      1.000    17268.3

   20     3584.0        -0.0    11887.3    -4693.9      1.000      1.000      1.000      1.000    18731.2
   30     3584.0        -0.0    34778.5     4693.9      1.000      1.000      1.000      1.000    39607.3

   30     3584.0        -0.0    29459.8    -9395.0      1.000      1.000      1.000      1.000    38526.0
   40     3584.0        -0.0    64303.5     9395.0      1.000      1.000      1.000      1.000    70576.6
```

The data fields change based on the piping code.

Multiple Piping Codes

If you analyzed the model with different piping codes, the software creates one stresses report for each code. Each report displays only the elements that were analyzed for the specific piping code.

The software filters the code compliance (pass or fail) check by piping code. If you want to see the compliance status of the entire model, run a Stresses Extended (Legacy) Report.

Offshore and Transportation Codes

The software determines code stress by the largest stress-to-allowable ratio.

ISO 14692

To ensure safe design, FRP pipe (ISO 14692 2005 and ISO 14692 2017) limits axial and hoop stress pairs to those within the allowable failure envelope. ISO 14692 2017 extends the envelope to include negative-value compressive axial stress and includes stress variations related to local shell conditions (hoop bending and axial roping stress). For more information, see the code-specific note ISO-14692.

NOTE A negative ratio always means passing criteria.

General Computed Results

General Computed Results lists reports, such as input listings or hanger selection reports, which are not associated with load cases.

Hanger Table and Hanger Table with Text

The **Hanger Table** report provides basic information regarding spring hangers either selected by CAESAR II or by you:

- Node number
- Number of springs required
- Hanger table figure number and size
- Hot load
- Theoretical installed load (the hanger setting in the field prior to pulling the pins)
- Actual installed load (the load on the hanger when the pipe is empty)
- Spring rate from the catalog
- Horizontal movement determined from the CAESAR II output.
- Hanger constant effort force (provided when constant effort supports are selected)

The **Hanger Table with Text** report displays more descriptive text and includes additional information:

- Variable support spring designed
- Maximum and minimum allowed single spring load
- Recommended installation clearance as read from the catalogs

Input Echo

Displays the **Input Listing Options** dialog to select which portions of the input are reported in this output format. All basic element data (geometry), operating conditions, material properties, boundary conditions, and report formatting are available:

- Elements
- Allowables
- Material ID
- Node Names
- Offsets
- Forces
- Uniform Loads
- Wind/Wave
- SIF's and TEE's
- Bends
- Rigids
- Expansion Joints
- Reducers
- Flanges
- Equipment Check
- Restraints
- Displacements
- Hanger
- Flexible Nozzles
- Units
- Coordinates
- Setup File
- Title
- Control Parameters

Select the options you want to print or view, and then select **OK**.

Miscellaneous Data

Displays the **Miscellaneous Data Options** dialog and allows you to select which portions of the results or of the output are reported in **Miscellaneous Report**. Select the options to print or view, and then select **OK** to display the report.

The following report options are available:

Bends SIF

SIF and flexibility data for bends.

Allowables

Allowable stress summary.

Tees SIF

SIF and flexibility data for tees.

NOTE B31J Methods provides directional stress intensification factors and flexibility factors. The software uses the highest in-plane or out-of-plane SIF when the piping code defines a stress equation that uses a single SIF. The software uses the flexibility factors for the specified orientation. See B31J Methods.

Reducers

Reducers report.

Nozzles

Nozzle flexibility data.

B31.3 MDMT

Minimum design metal temperature analysis for the brittle fracture of piping according to B31.3.

Pipe Prop #1

Pipe properties report with weights and minimum calculated wall thickness for each element.

Pipe Prop #2

Pipe properties report with weights and minimum calculated wall thickness for each element.

C.G.

Center-of-gravity report.

B.O.M.

Bill of materials.

Wind

Wind input data.

Wave

Wave input data.

Seismic

Seismic input and calculated g factors (magnifiers of gravitational loading).

B31.8 Buckling Check

Stress and strain values to determine buckling and lateral instability for a pipe element according to B31.8-2018 paragraph 833.10.

Load Case Report

The **Load Case Report** documents the Basic Names (as built in the Load Case Builder), User-Defined Names, Combination Methods, Load Cycles, and Load Cases (Output Status, Output Type, Snubber Status, Hanger Stiffness Status, and Friction Multiplier) of the static load cases. This report is available from the **General Computed Results** column of the **Static Output Processor**.

```
CAESAR II
Job: C:\TEMP\CAESAR II\MYFIRSTPIPINGMODEL
Licensed To: Seat   --   ID #17253

 CAESAR II  LOAD CASE REPORT

CASE 1 (OPE) W+T1+P1
OPERATING CASE CONDITION 1
     Keep/Discard:          Keep
     Display:               Disp/Force/Stress
     Elastic Modulus:       EC
     Friction Mult.:        1.0000
     Flg Analysis Temp:     None

CASE 2 (SUS) W+P1
SUSTAINED CASE CONDITION 1
     Keep/Discard:          Keep
     Display:               Disp/Force/Stress
     Elastic Modulus:       EC
     Friction Mult.:        1.0000
     Flg Analysis Temp:     None

CASE 3 (EXP) L3=L1-L2
EXPANSION CASE CONDITION 1
     Keep/Discard:          Keep
     Display:               Disp/Force/Stress
     Combination Method:    ALG
     Flg Analysis Temp:     None
```

Warnings

All warnings reported during the error checking process are summarized here.

```
CAESAR II
Job Name: 1001-P
Licensed To: SPLM: Edit company name in <system>\company.txt

 CAESAR II  WARNINGS REPORT (FROM ERROR CHECKER)


    WARNING 169E   On element    20 TO    30 the second reducer
    thickness was not specified.  CAESAR II will use
    a default value of   9.271.

    WARNING 172E   On element    20 TO    30 the reducer alpha
    value was not specified.  CAESAR II will use
    a default value of:    14.197.
```

Output Viewer Wizard

Displays options for creating output reports.

Report Formats

Send to Screen

Displays combined reports in a tabbed window. The software does not save an output file. See On Screen.

Send to MS Word

Displays combined reports in a .docx file. See Using Microsoft Word.

Send to Printer

Sends the reports directly to the selected printer. The software does not save an output file.

Send to Text (ASCIII) File

Saves a combined report text file to the selected location.

Send to MS Excel

Displays combined reports in a tabbed .xlsx file. See Using Microsoft Excel.

Send to PDF

Displays combined reports in a .pdf file and saves the file to the selected location. See Using PDF.

Generate Table of Contents (TOC)

Creates a table of contents of the created reports.

NOTE The software includes a list of the static load cases with the table of contents.

Show these reports in this order

Displays a list of the added reports in their output order.

→ **Add**

Adds selected reports from Standard Reports and General Computed Results to the reports list.

NOTES

- Before adding a report, also select the load cases to include in the report from Load Cases Analyzed.
- Hold the SHIFT or CTRL key to select more than one load case or report.

← **Remove**

Removes a selected report from the reports list.

Clear All

Clears all reports from the reports list.

Move Up / Move Down

Moves a selected report up or down in the order on the reports list.

Finish

Creates the reports in the reports list using the select report format.

Report Template Editor

Static Output Processor menu: **Options** > **Custom Reports** > **New**

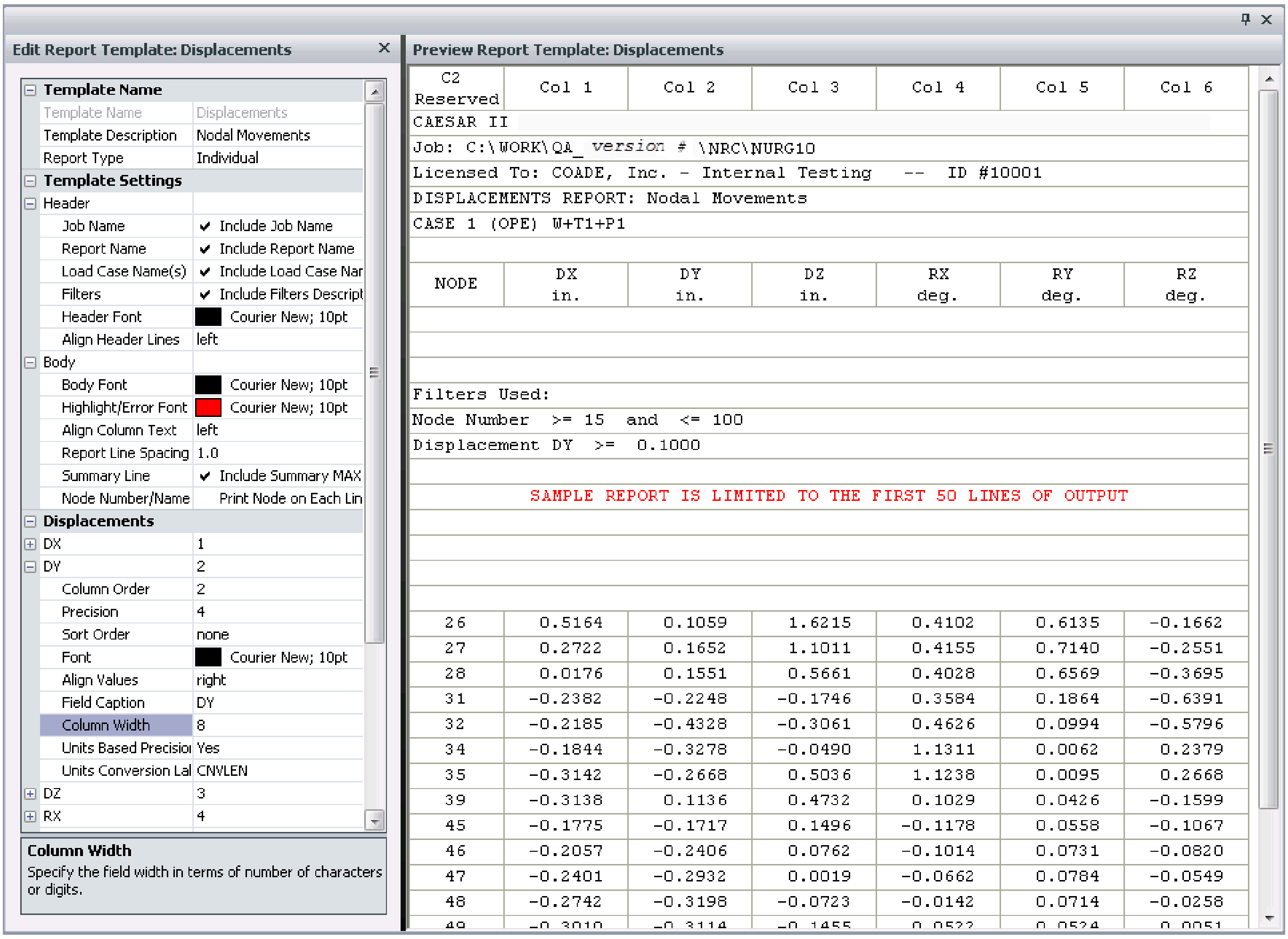
Custom Reports toolbar: **Add new custom report template**

Creates custom reports using the **Report Template Editor**. To create a new report, first select one or more load cases, and then select the command.

Static Output Processor menu: **Options** > **Custom Reports** > **Edit**

Custom Reports toolbar: **Edit an existing custom report template**

Edits existing reports using the **Report Template Editor**. To customize an existing report, select the load case, select a standard or custom report name, and then select the command.



The **Report Template Editor** dialog consists of two sections: the template editor to the left and the preview grid to the right.

The template editor has a tree-like structure and resembles Window Explorer's folder view. There are 11 major categories available: Template Name and Template Settings for general report editing, and several output fields; Displacements, Restraints, Local Restraints, Equipment Nozzle Checks, Global and Local Forces, Flange Evaluation, Stresses, and Hanger Table Data.

The **Template Name** category allows you to specify the report name, enter a brief description of the report, and select the report type. The report name followed by the template description displays on the preview grid if the **Include Report Name** option is checked under the **Template Settings** category.

There are three report types available:

- **Individual**

Generates output reports, one per selected load case, in a format similar to the standard **Displacements** or **Restraints** reports.

- **Summary**

Generates a single output report for all the specified load cases as a summary, in a format similar to the standard **Restraint Summary** report.

- **Code Compliance**

Generates an output stress check report for multiple load cases as a single report, similar to the standard **Code Compliance** report.

NOTE Actual columns and their order on the reports are controlled solely by you. Data from various categories can be customized on a single report to suit your needs.

The **Template Settings** category provides options for the report header and the report body text, formatting, and alignment. Also set the font face, size, and color for the header and the report body. Include or remove specific header text (such as **Report Name**, **Job Title** or **Filters Description**) by selecting and clearing the checkbox next to the corresponding item. **Report Line Spacing** changes the spacing between lines of text. The **Summary Line** option (used with Summary-type reports) toggles the appearance of the summary line with MAX values for each field or column per node. Select the **Node Number/Name** option (used with Summary-type reports) to repeat the Node information on each **Loadcase** line. If you clear this option, then the node will appear on the separate line above the data for load cases. These two options may help with later data manipulations when sending the reports to a Microsoft Excel spreadsheet

NOTE Any changes in the editor are immediately reflected in the preview window.

Each of the following categories consists of related output data. For example, the **Displacements** category contains three translational (**DX**, **DY**, and **DZ**) and three rotational (**RX**, **RY**, and **RZ**) fields, **Stresses** contains **Axial**, **Bending**, and **Code** stresses among other stress related fields. A number next to the field name indicates the Column Order this field will be placed in. When nothing or a zero value is specified, this column will not be included in the current report.

Each field contains the following information:

Field Name	Description
Column Number	Indicates the order of the fields in the output report.
Precision	Indicates the number of decimal places to be displayed.
Sort Order	Specifies whether the data in the column is in ascending, descending, or in no order. This gives you flexibility of reviewing reports for maximum (or minimum) values.

Font	Specifies the text font face, size and color for this field whenever special formatting is required. Set the generic font settings for the entire report at the **Template Settings > Body** category.
Align Values	Controls left, right, or center alignment of the values in the column.
Field Caption	Customize the name of the field as it appears on the report. This may be useful to customize the display of the output displacements in the report to reflect the plant North/South/East/West directions or vertical and horizontal notations instead of generic X, Y, Z.
Column Width	Controls the size of the column in terms of the number of displayed characters or digits. In addition, resizing the columns in the **Preview Grid** adjusts the **Column Width** value. Type **0** to close the column and remove it from the report. Type **-1** to size the column to the predefined default size.
Units Based Precision	Indicates whether to enable the automatic control of the displayed number of decimal places to be calculated based on the selected display units. This value is used together with the **Units Conversion Label** value. The **Precision** value is ignored in this case. When set to **No**, the **Precision** value takes place.

NOTE When a category or any field is highlighted in the editor, the help text for this field displays in the **Help** box at the bottom of the editor window.

The **Preview Grid** on the right of the **Custom Report Template Editor** dialog is interactive. Drag the columns by their heading to arrange the order of the fields in the reports. Double-clicking the column header sorts that column's values in ascending or descending order. The dragged column number or sorted order value will automatically be saved in the **Column Number** or **Sort Order** entry of that field in the editor tree. Select the column header once to highlight that field in the editor tree, extend its contents and scroll it to view.

NOTE The **Preview Grid** is limited to the first 50 lines. The entire report is available after you select the appropriate load cases and custom report name on the **Static Output Processor** dialog and select **View Report**.

Any current changes to the custom report template can be saved by selecting **Save**. The custom report template can also be saved under a different name by selecting **Save As...** The **Save As...** dialog prompts you to enter the new template name, a brief description, and the report type. Select **Preview Report** to remove the grid lines from the **Preview Grid**. Select the button again to add the grid lines for editing.

Available Commands

The **Static Output Processor** window menus and toolbars provide commands to review, create, and modify reports. The **3D/HOOPS Graphics** toolbars navigate and display report information in graphics mode.

View Menu

Activates and disables toolbars.

Standard Toolbar

Open

Opens a different job for output review. You are prompted for the file to open.

Save

Saves the selected reports to a text file. You are prompted for the file name. A table of contents for all currently selected reports is added to the end of the text file.

Load Case Name

Selects either the **CAESAR II Default Load Case Names** or the **User-Defined Load Case Names** for output reports. The selected name also displays in the **Load Cases Analyzed** box in the **Static Output Processor** window. You enter user-defined load case names in the Load Cases Tab (Static Analysis - Load Case Editor Dialog).

Node Name

Defines the formatting of the node numbers and names for generated reports. See Node Name.

Title Lines

Inserts report titles for a group of reports. See Title Lines.

Return to Input

Opens Piping Input.

View Animation

Shows animation of the displacement solution. See View Animation.

Graphical Output

Superimposes analytical results onto a plot of the system model. See Graphical Output.

Print

Prints the selected reports. The software does not save an output file.

View report using Microsoft Word

Send the report directly to Microsoft Word. See Using Microsoft Word.

View report using Microsoft Excel

Sends output reports directly to Excel. See Using Microsoft Excel.

View report using PDF

Sends output reports to a .pdf file. See Using PDF.

On Screen

Displays the selected reports in a window on the computer screen. See On Screen.

Displacements Toolbar

Maximum Displacements

Places the actual magnitude of the X, Y, or Z displacements on the currently displayed model.

NOTE The element containing the displaced node is highlighted, and the camera viewpoint is repositioned preserving the optical distance to the model. This brings the displaced node to the center of the view.

a. The software starts with the highest value for the given direction. After you press **Enter**, the remaining values are placed in a similar manner until all values become zero.

b. Select **Maximum Displacements** again to clear the view of the displayed values and highlighting.

TIP Select **Show** > **Displacement** > **Maximum Displacement** >**X, Y,** or **Z** to access this command from the menu. If **Show Element Viewer Grid** is selected, then the viewer displays the Displacements report for the selected load case and highlights the column and row to represent the displacement direction and current node.

Grow Toolbar

Deflected Shape

Overlays the scaled geometry with a different color into the current plot for the selected load case. Select the down arrow to display an additional menu with the selected feature checked and the **Adjust Deflection Scale** option.

Adjust Deflection Scale

Specifies the deflected shape plot scale factor. You may not be able to see the deflected shape if the value is too small. If you enter a scale value that is too large, the model may be discontinued. Select **Show** > **Displacement** > **Scale** to access this command from the menu.

Grow

Displays the expansion of a selected pipe due to the addition of heat.

Restraints Toolbar

Output Restraints Symbols

Adds restraint symbols to the plot. Restraints are plotted as arrowheads with the direction of the arrow indicating the direction of the force exerted by the restraint on the piping geometry.

Maximum Restraint Loads

Places the actual magnitude of the calculated restraint loads for a selected load case on the currently displayed geometry. **Maximum Restraints Loads** displays the load magnitude value next to the node, highlights the element containing the node, and is brought to the center of the graphics view. The **Zoom to**

Selection and **Show EventViewer Grid** options are still available. After pressing **Enter**, any remaining values are placed in a similar manner.

Stresses Toolbar

Overstress

Displays the overstressed point distribution for a load case. Nodes with a calculated **code stress to allowable stress ratio** of 100% or more display in red. The remaining nodes or elements display in the color selected for the lowest percent ratio. This feature is useful to quickly observe the overstressed areas in the model.

Overstressed conditions are only detected for load cases where a code compliance check was done (such as where there are allowable stresses available).

Overstressed nodes display in red in the **Event Viewer** dialog (if it is enabled).

The model is still fully functional. Zoom, pan, or rotate it.

Maximum Code Stress

Displays the stress magnitudes in descending order.

NOTE **Maximum Code Stress** operation is similar to **Maximum Displacements.** The stress value is displayed next to the node, and the element containing the node is highlighted and moved to the center of the view.

If needed, use the **Zoom to Selection** and **Show Event Viewer Grid** options. Press **Enter** and the next highest value is placed with corresponding element highlighting.

In addition to the numbers that could be found in a corresponding report, this command provides a graphical representation and distribution of large, calculated code stresses throughout the system.

Code Stress Colors by Value

Displays the piping system in a range of colors where the color corresponds to a certain boundary value of the code stress. Use this feature to see the distribution of the code stresses in the model for a load case.

In addition to the model color highlight in the graphics view, the corresponding color key legend window is displayed in the top left corner of the graphics view. The legend window can be resized and moved.

The colors and corresponding stress levels can be set in the **Configuration/Environment**. For more information, see Configuration Editor.

Code Stress Colors by Percent

Displays the piping system in a range of colors, where the color corresponds to a certain percentage ratio of code stress to allowable stress. This option is only valid for load cases where a code compliance check was done such as where there are allowable stresses.

Us**e Code Stress Colors by Percent** to see the distribution of the code stress to allowable ratios in the model for a load case. The legend window with the corresponding color key also displays in the upper-left corner of the graphics view. The legend window can be resized and moved.

Selecting the arrow to the right of the button displays an additional menu with two options: **Display** and **Adjust Settings**. Selecting the **Display** option displays the color distribution. Selecting the **Adjust Settings** option displays the color distribution. Selecting the **Adjust Settings** option displays the **Stress Settings** dialog where values and corresponding colors can be set or adjusted. These settings are related to the job for which they are set. The settings are saved in the corresponding *job_name*.XML file in the current job data directory see Customize the graphics.

Reports Navigation Toolbar

Navigation commands in this toolbar become enabled by selecting at least one report.

← / → View Previous Report / View Next Report

Navigates through the report tabs.

Go To

Displays the list of currently-opened reports in alphabetical order.

Find in Report

Provides search capabilities for a specific node number, maximum values of any of the report fields, of for any text or number.

+ / – Zoom In / Zoom Out

Zooms the view in or out without affecting the actual report font or formatting. The zoom level can also be controlled from the right-click context menu. The zoom level is applied to the current report and is temporal until the report is closed.

Save Current Custom Report Template

Saves the changes to the custom report when the **Report Template Editor** is opened.

Save Current Custom Report Template with a New Name

Enables keeping the original report and saving the changes to another report when the **Report Template Editor** is launched.

Preview Report

Removes the grid lines from the **Preview Grid**. Selecting the button again adds the grid lines.

Custom Reports Toolbar

Commands in the **Custom Reports** toolbar enable you to manipulate the generated reports.

Add New Custom Report Template

Creates a new custom report. For more information, see [New Custom Report Template].

Edit Existing Custom Report Template

Modifies an existing custom report. For more information, see [Edit Custom Report Template].

Delete Custom Report Template

Deletes a custom report. For more information, see [Delete Custom Report Template].

Reset Default Custom Report Templates

Replaces the current custom report templates with the default templates. For more information, see Reset Default Custom Report Templates.

Import Custom Report

Imports a custom report template. For more information, see Import Custom Report.

Export Custom Report

Saves any custom generated report to a text file. For more information, see Export Custom Report.

Options Menu

Specifies common settings that are available on all reports such as how node numbers display and title information.

On Screen

Static Output Processor menu: **Options** > **View Reports** > **On Screen**

Standard toolbar: **View Reports**

Output Viewer Wizard: **Send to Screen**

Displays the selected reports in text format.

Multiple load cases and reports can be sent. Each report is presented one at a time for inspection.

Set Report Font

Static Output Processor menu: **Options** > **View Reports** > **On Set Report Font**

Standard toolbar: **View Reports** > **Set Report Font**

Opens the **Font** dialog used to define the text font, font style, and font size.

NOTE Some screen fonts may not be available on your printer. If the font is not available for your printer, the closest matching font on your printer is used.

Using Microsoft Word

Static Output Processor menu: **Options** > **View Reports** > **Using Microsoft Word**

Standard toolbar: **View report using Microsoft Word**

Output Viewer Wizard: **Send to MS Word**

Sends output reports directly to Microsoft Word. Append additional reports to form a final report.

Word is available as an output device to the **Static Output Processor** and the **Dynamic Output** windows.

1. Select the required load cases and reports.
2. Select **View report Using Microsoft Word**.
3. Repeat steps 1 and 2 to append additional reports to the Word document.

To create a table of contents on the report, select **Generate Table of Contents (TOC)** on the Output Viewer Wizard before creating the report.

Using Microsoft Excel

Static Output Processor menu: **Options** > **View Reports** > **Using Microsoft Excel**

Standard toolbar: **View report using Microsoft Excel**

Output Viewer Wizard: **Send to MS Excel**

Sends output reports directly to Excel. Append additional reports to form a final report.

1. Select the required load cases and reports.
2. Select **View report using Microsoft Excel**.
3. Repeat steps 1 and 2 to append additional reports to the Excel document.

To create a table of contents on the report, select **Generate Table of Contents (TOC)** on the Output Viewer Wizard before creating the report.

Using PDF

Static Output Processor menu: **Options** > **View Reports** > **Using PDF**

Standard toolbar: **View report using PDF**

Output Viewer Wizard: **Send to PDF**

Sends output reports directly to a .pdf file. Multiple load cases and reports can be sent to a single .pdf file.

Reports cannot be appended to a .pdf file like with Using Microsoft Word and Using Microsoft Excel. Each selection of **Using PDF** creates a new report.

To create a table of contents on the report, select **Generate Table of Contents (TOC)** on the Output Viewer Wizard before creating the report.

New Custom Report Template

Static Output Processor menu: **Options** > **Custom Reports** > **New**

Custom Reports toolbar: **Add new custom report template**

Creates a new custom report using the Report Template Editor dialog. Select at least one load case from the **Load Cases Analyzed** list before creating a new report template.

1. From the **Load Cases Analyzed** list, select the load case for the custom report template.

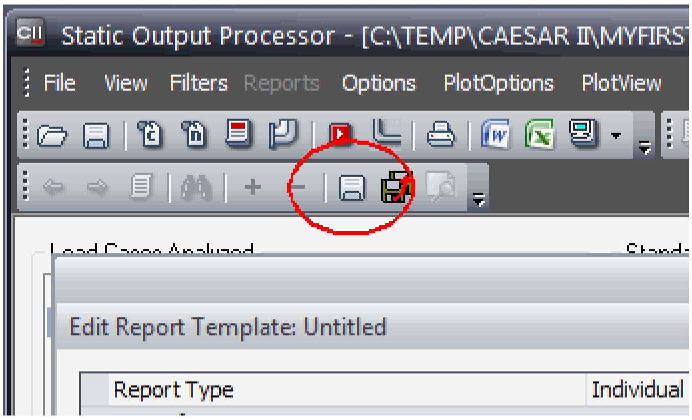
2. Select **Options > Custom Reports > New**.

3. In the **Template Name** box, enter a name for your custom report.

4. In the **Template Description** box, enter a description.

5. Using the Report Template Editor options, create your custom report.

6. Select **Save Current Custom Report Template** on the **Reports Navigation** toolbar.



TIP Do not use **File > Save** or the **Save** command on the main toolbar.

*Your report appears in the **Custom Reports** list.*

Edit Custom Report Template

Static Output Processor menu: **Options** > **Custom Reports** > **Edit**

Custom Reports toolbar: **Edit an existing custom report template**

Modifies and saves existing custom reports using the Report Template Editor.

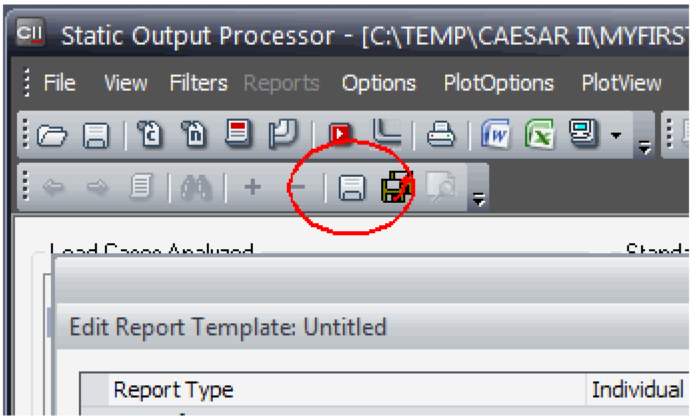
1. Select one or more load cases from the list.

2. From the **Custom Reports** list, select the report to edit.

3. Select **Options > Custom Reports > Edit**.

4. Using the **Report Template Editor** dialog options, edit your custom report.

5. Select **Save Current Custom Report Template** on the **Reports Navigation** toolbar.



TIP Do not use **File > Save** or the **Save** command on the main toolbar.

- OR -

Select **Save Current Custom Report Template with a New Name** to save your edit to a new custom report leaving the original report unchanged.

Delete Custom Report Template

Static Output Processor menu: **Options** > **Custom Reports** > **Delete**

Custom Reports toolbar: **Delete one or more custom report templates**

Deletes a custom report template. Standard delivered reports cannot be deleted using this command.

IMPORTANT Undo is not available for the deletion of a custom report template.

1. From the **Custom Reports** list, select the report to delete.
2. Select **Options > Custom Reports > Delete** .
3. Select **Yes** to confirm that you want to delete the report.

Reset Default Custom Report Templates

Static Output Processor menu: **Options** > **Custom Reports** > **Reset**

Custom Reports toolbar: **Reset to CAESAR II default custom report templates**

Replaces the current report templates, both CAESAR II delivered and custom defined reports, with the default report templates delivered with the software. Use this command if you received a new version or a patch of CAESAR II and want to use the new reports.

IMPORTANT Make sure that you export any custom reports that you want to keep before using this command. This command affects ALL jobs system-wide and cannot be undone. For more information about exporting custom reports, see Export Custom Report.

Import Custom Report

Static Output Processor menu: **Options** > **Custom Reports** > **Import**

Custom Reports toolbar: **Import a custom report**

Imports a custom report template that was exported earlier using **Export**.

The report template file extension is *.C2RPT and can be read from any network location. After the report template file is imported, it becomes a part of the current configuration. The new report is appended to the **Custom Reports** list of the **Static Output Processor** window. The default name of the template file corresponds to the custom report name.

Export Custom Report

Static Output Processor menu: **Options** > **Custom Reports** > **Export**

Custom Reports toolbar: **Export a custom report**

Saves any custom generated report to a text file. The report template file name extension is *.C2RPT and can be saved to any accessible location. The default file name is the custom report name. Use **Import** to import these saved custom reports.

1. In the **Custom Reports** list, select the report to export.
2. Select **Options** > **Custom Reports > Export**.
3. Select a folder and enter a file name.
4. Select **Save**.

View Animation

Main window ribbon: **Output** > **Animations** > **Static**

Static Output Processor menu: **Options** > **View Animation**

Static Output Processor Standard toolbar: **View Animation**

Shortcut key:

- **ALT+M - Motion**

Displays the piping system as it moves to the displaced position of the basic load cases. To animate the static results, select **Options** > **View Animations** to display the **DynPlot** window.

The **Animated Plot** menu has several plot selections. **Motion** and **Volume Motion** are the commands to activate the animation. **Motion** uses centerline representation while **Volume Motion** produces 3D graphics. Select the load case from the drop-down list. Animations may be sped up, slowed down, or stopped using the toolbars.

CAESAR II also enables you to save animated plots as HTML files by selecting **File** > **Save As Animation**. After saving these files, view them on any computer outside of CAESAR II.

NOTE The corresponding animation graphics file <*job_name*>.hsf must be transferred along with the HTML file for proper display.

Animation of Static Results -Displacements

View the piping system as it moves to the displaced position for the basic load cases. To animate the static results, select **Static Output** > **Options** > **View Animation**.

TIP Select **View Animation** to view graphic animation of the displacement solution.

Static animation graphics has all the standard model projection and motion toolbar commands. The load case can be selected from the drop-down list. The title consists of the load case name followed by the file name and can be toggled on and off from the **Action** menu.

The **Static Animation** processor allows viewing of the single line and volume motion, controls the speed of the movement, and the animation can be saved to a file as described above.

NOTE We recommend you use the **Deflected Shape** command button on the 3D/HOOPS Graphics view of the **Static Output Processor** toolbar. For more information refer to **3D/HOOPS Graphics Tutorial for Static Output Processor, Deflected Shape**.

Graphical Output

Static Output Processor menu: **Options** > **Graphical Output**

Standard toolbar: **3D Plot**

To support a graphics mode, the **Static Output Processor** window provides 3D/HOOPS Graphics toolbars that contain commands to zoom, orbit, and pan, as well as provide the ability to switch views and modes.

The 3D/HOOPS Graphics toolbar commands include the display of displaced shapes, highlighting and zooming to maximum displacements, restraint loads, and stresses of the model. Another advantage provided by 3D/HOOPS graphics is the graphical representation of stresses by value and by percentage use color.

A variety of CAESAR II Output Plot functions, accessed from the 3D/HOOPS Graphics toolbars or the **Show** menu, are broken into submenus:

- Displacements
- Restraints
- Forces/Moments
- Stresses

Selection Options Toolbar

Show Event Viewer Grid

Shows or hides the **Event Viewer** on the plot. See Element Viewer Dialog.

Zoom to Selection

Fits the selected element in the view.

Organization Tools Toolbar

Line Numbers

Displays the Line Numbers Dialog, which allows graphical editing of line numbers. Perform the following from this pane.

- Assign a new line number to the block of elements that have been selected on the 3D graphical display.
- Remove an existing line number.
- Set and reset visibility options to hide and unhide elements.
- Assign a color to an individual line number.

To reassign one or more elements from one line number to another existing line number, simply drag-and-drop (move) the elements between existing Line Numbers in the Line Numbers Dialog.

When you select the **Line Number** name in the Line Numbers Dialog, the corresponding elements are highlighted in the 3D pane and are selected to perform block (global) operations.

Line Numbers Dialog

Controls options of the line number or name for a pipeline/pipe run containing one or more pipe elements. Set options for line numbers on the Classic Piping Input Dialog and the Static Output Processor.

Create (Create from Selection)

Creates a pipe run from the selected elements. Select elements from the 3D model or the **Line Numbers** dialog. The line number is given the default name *Line Number <x>*, where *<x>* is a sequential number. This option is only available in the Classic Piping Input Dialog**.**

Remove (Remove Line Number)

Deletes the selected pipe run line numbers. Elements in the pipeline move to the next line number up in the sequence. This option is only available in thr Classic Piping Input Dialog**.**

NOTE Alternatively, right-click and select **Remove Line Number**.

Reset (Reset Settings)

Returns settings for all line numbers and their elements to their default values. Use the drop down to select **Reset Visibility**, **Reset Color**, or **Reset All**.

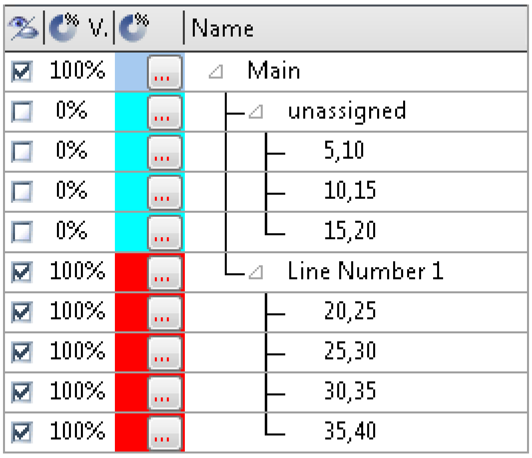
NOTE Alternatively, right-click on a line and select **Reset Visibility**, **Reset Color**, or **Reset All** to only reset the selected line number.

Line Number Views

Line number and element rows display in a tree view. Elements are named by their beginning and ending node numbers. Create a view using the following methods:

- Select a row to change the visibility to 100%. Clear a row to change the visibility to 0%.
- When you select or clear a line number, the software also selects or clears all of line number's elements. Then select or clear individual elements.
- Select or clear **Main** to change the selection of all line numbers and elements. Then select or clear individual line numbers and elements.

NOTE Press SHIFT + click to select multiple line numbers or multiple elements.



Previous (Previous View)

Saves the current view and returns to the previous view. If no view is saved, all rows are selected.

NOTE This option is not available in the Static Output Processor.

Invert (Invert Selection)

Reverses the line number selection to clear the selection of previously-selected rows and to select the rows not previously selected.

NOTE This option is not available in the Static Output Processor.

<type here to search>

Limits the elements that display in the **Line Numbers** dialog to pipe run line numbers or elements that match the text in this field. Clear this field to display all line numbers and elements in the tree view. Search for a name or a node number.

Show/Hide

Turns the display of line numbers and elements on or off. Clear a line number to reduce visibility to 0% for the line number and its elements. Clear an element to reduce visibility to 0% for only that element.

NOTE If node numbers are turned on, node numbers do not display when the element opacity is 0%. For more information on displaying node numbers, see Node Numbers.

V. **Visibility**

Specifies the opacity of line numbers and elements. 100% indicates that the element is opaque. 0% indicates that the element is invisible. Specify the value of a line number to change opacity for the line number and its elements. Specify the value of an element to change opacity for only that element.

NOTE If node numbers are turned on, the node number opacity matches the element opacity. For more information on displaying node numbers, see Node Numbers.

Color

Displays the **Color** dialog to specify a color for a line number and its elements.

Name

Displays the name of line number and elements. Select a line number to change its name.

New Analysis Reviewer

Main window ribbon: **Home** > **New Analysis Reviewer**

Main window ribbon: **Output** > **Reports** > **New Analysis Reviewer**

In the **Classic Piping Input** window:

CAESAR II Tools toolbar: **View Static Results**

In the **Static Output Processor** window:

Options menu > **New Analysis Reviewer**

Standard toolbar: **New Analysis Reviewer**

Opens the New Analysis Reviewer, which provides advanced, flexible options for reviewing output results and building reports.

NOTE This command is not available when **Classic Output Processor** is selected for Preferred Output Reviewer in the Configuration Editor.

Data Export Wizard

Main window Interfaces tab: **Generic** > **Data Export Wizard**

Static Output Processor menu: **Options** > **Data Export Wizard**

Static Output Processor Standard toolbar: **Data Export Wizard**

Provides export of both the input model and output data. For more information, see Data Export Wizard.

After finishing an export or canceling the **Data Export Wizard**, the software returns to the **Static Output Processor**.

Title Lines

Static Output Processor menu: **Options** > **Title Lines**

Standard toolbar: **Enter Report Titles**

Inserts report titles for a group of reports. Define a two-line title or description for a report. The title can be assigned once for all load case reports sent to the printer or a disk drive; or the title can be changed for each individual report before it is moved to the output device.

The title line allows for 28 characters per line.

Load Case Name

Static Output Processor menu: **Options** > **Load Case Name**

Standard toolbar: **Select Case Names**

Selects either the **CAESAR II Default Load Case Names** or the **User-Defined Load Case Names** for output reports. The selected name also displays in the **Load Cases Analyzed** box in the **Static Output Processor** window. You enter user-defined load case names in the Load Cases Tab (Static Analysis - Load Case Editor Dialog).

Node Name

Static Output Processor menu: **Options** > **Node Name**

Standard toolbar: **Select Node Names**

Defines the formatting of the node numbers and names for generated reports.

Select the format to use from the **Node Name Choice** dialog:

- Node Number Only
- Number (Name)
- Number -- Name
- Node Name Only
- Name (Number)
- Name -- (Number)

Return to Input

Static Output Processor menu: **Options** > **Return to Input**

Standard toolbar: **Back to Input**

Opens Piping Input.

Plot Options Menu

Performs actions associated with the display of the model. **Plot Options** commands area available after you select **Graphical Output**.

Range

Piping Input menu: **Options** > **Range**

Piping Input Plot Tools toolbar: **Range**

Static Output Processor menu: **PlotOptions** > **Range**

Static Output Processor Plot Tools toolbar: **Range**

Shortcut key: **CTRL**+**ALT**+**U**

Displays only the elements that contain nodes within a range. This is helpful when you need to locate specific nodes or a group of related elements in a large model. This command displays the Range dialog.

NOTES

- Using **Range** affects the display and operation of other 3D graphics highlighting options. For example:
    - If part of the model is not visible because of **Range**, then the Diameters command only highlights the elements that are visible.
    - If **Range** hides any nodes containing the predefined displacements, the Displacements legend grid still displays, but the model may not highlight correctly.
- **Find** may not work properly for the part of the model that is hidden by the range. The corresponding message displays in the status bar.

Range Dialog

Range Dialog

Controls options for manipulating ranges.

Show only

Specifies the items to show.

From

Specifies the node number for the start of the range.

To

Specifies the node number for the end of the range.

At These Elements

All elements that exist in the model are displayed. This list indicates which elements are included in the range. Clear this option for elements that are not included.

Add

Adds an item to the **At These Elements** list.

Reverse Selection

Clears all elements that were selected in the **At These Elements** list and selects all elements that were cleared.

Select All

Selects all elements in the **At These Elements** list.

Clear All

Clears all elements in the **At These Elements** list.

Restraints

Piping Input menu: **Options** > **Restraints**

Piping Input Plot Tools toolbar: **Restraints**

Static Output Processor menu: **PlotOptions** > **Restraints**

Static Output Processor Plot Tools toolbar: **Supports**

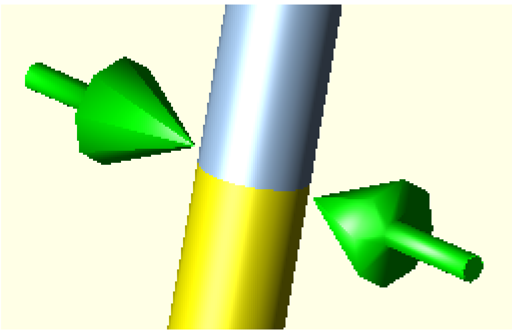
Shortcut keys: **CTRL**+**ALT**+**R**

Turns the display of restraints on or off on the current model.

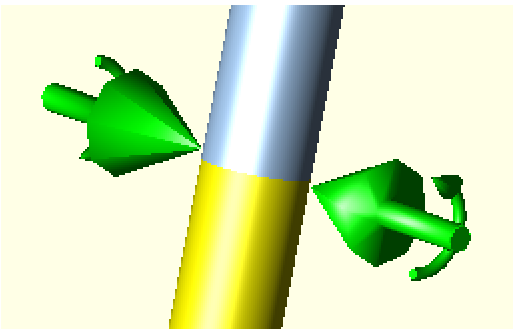
On the toolbar, select the arrow on the icon to indicate the display size of the restraints and whether the software displays restraints with or without connecting nodes (CNodes).

When **Restraints** is turned on the software displays:

- A directional arrow for a restraint.



- A directional arrow and curved arrow (following the right-hand rule) for rotational restraints, such as **RX**, **RY**, or **RZ**.



NOTE To graphically display restraint gaps, use the **Restraint** legend. For more information, see Legends Toolbar and Using Legends to Check Your Model.

Anchors

Piping Input menu: **Options** > **Anchors**

Piping Input Plot Tools toolbar: **Anchors**

Static Output Processor menu: **PlotOptions** > **Anchors**

Static Output Processor Plot Tools toolbar: **Anchors**

Shortcut key: **CTRL+ALT+A**

Turns the display of anchors on or off.

Select the arrow on the icon to indicate the anchor size to display on your model, as well as whether the software displays anchors with or without connecting nodes (CNodes).

Displacements

Piping Input menu: **Options** > **Displacements**

Piping Input Plot Tools toolbar: **Displacements**

Static Output Processor menu: **PlotOptions** > **Displacements**

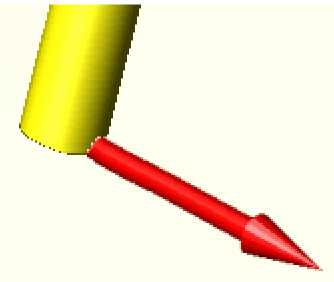
Static Output Processor Plot Tools toolbar: **Displacement**

Shortcut key: **CTRL+SHIFT+D**

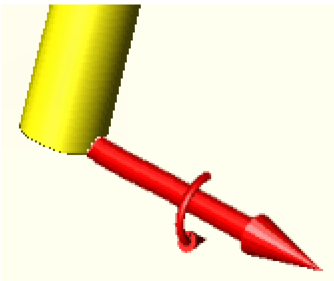
Turns the display of displacements on or off. This option also controls the display of displacements on CNode restraints.

When **Displacements** is turned on, the software displays:

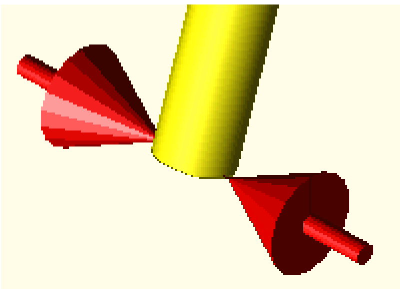
- A directional arrow for the resultant linear displacement vector.



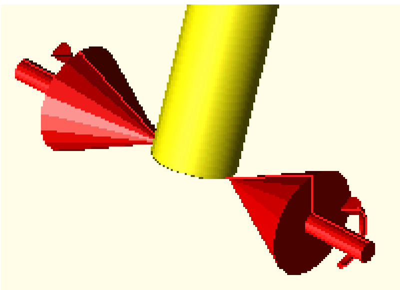
- A directional arrow and curved arrow (following the right-hand rule) for the resultant rotational displacement vector.



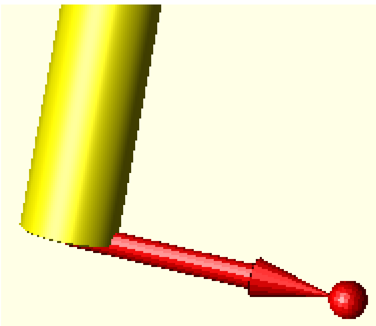
- A pair of shorter directional arrows for **Fixed** linear displacement. (Disp. Value = 0)



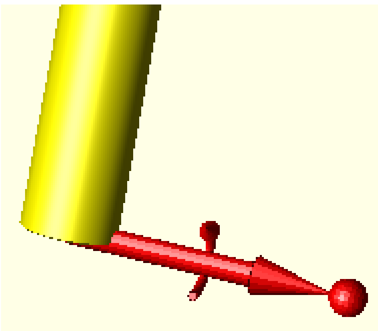
- A pair of shorter directional arrows with curved arrows for **Fixed** rotational displacement.



- A directional arrow with a sphere at the top for non-fixed displacement which indicates hidden fixed vectors.



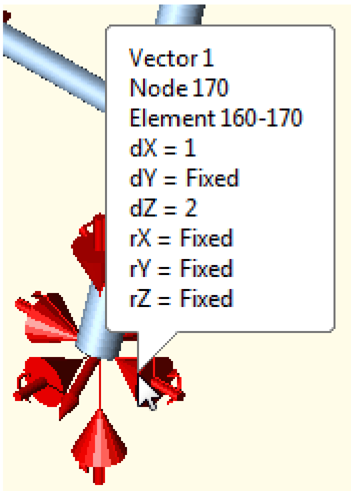
- A directional arrow and curved arrow (following the right-hand rule) with a sphere at the top for the resultant rotational non-fixed displacement which indicates hidden fixed vectors.



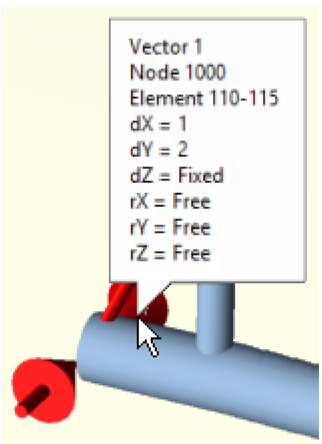
Other specifications:

- The arrow size - **Largest**, **Larger**, **Medium**, **Smaller**, or **Smallest**.
- Whether to hide or show fixed displacements - **Show Fixed**.
- The vector to display - **Vector1**, **Vector2**, and so on.

Hover the cursor over displacement arrows to see the displacement values for the displayed vector:



Where a restraint has a CNode with displacement, the displacements are displayed with values displayed with the Node number:



NOTE Change the default arrow colors in the Graphic Settings of the Configuration Editor or by using **Plot Properties**. For more information, see Displacements and Display Options Toolbar.

Hangers

Piping Input menu: **Options** > **Hangers**

Piping Input Plot Tools toolbar: **Hangers**

Static Output Processor menu: **PlotOptions** > **Hangers**

Static Output Processor Plot Tools toolbar: **Hangers**

Shortcut key: **CTRL+ALT+H**

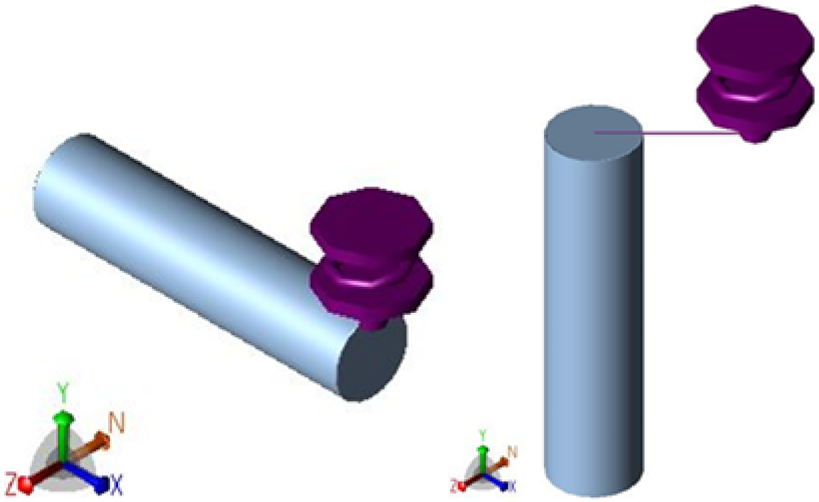
Turns the display of hangers and cans on or off.

NOTE This is a graphical representation of the number of hangers at the location, not of the hanger installation.

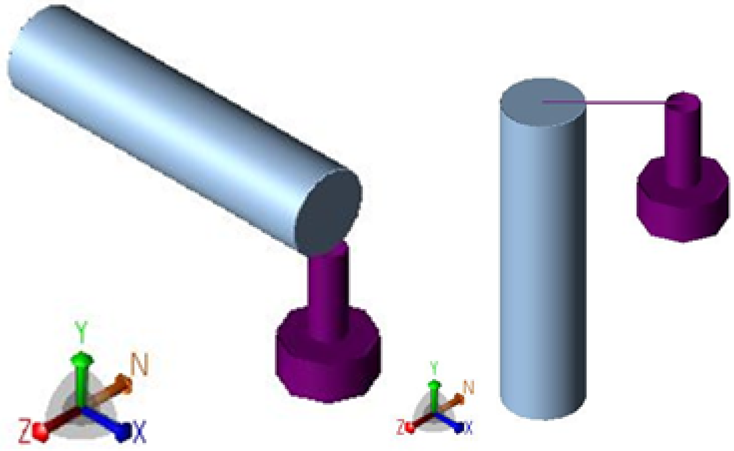
On the toolbar, select the arrow on the icon to indicate the display size of the hangers and whether the software displays hangers with or without connecting nodes (CNodes).

When **Hangers** is turned on, the software displays:

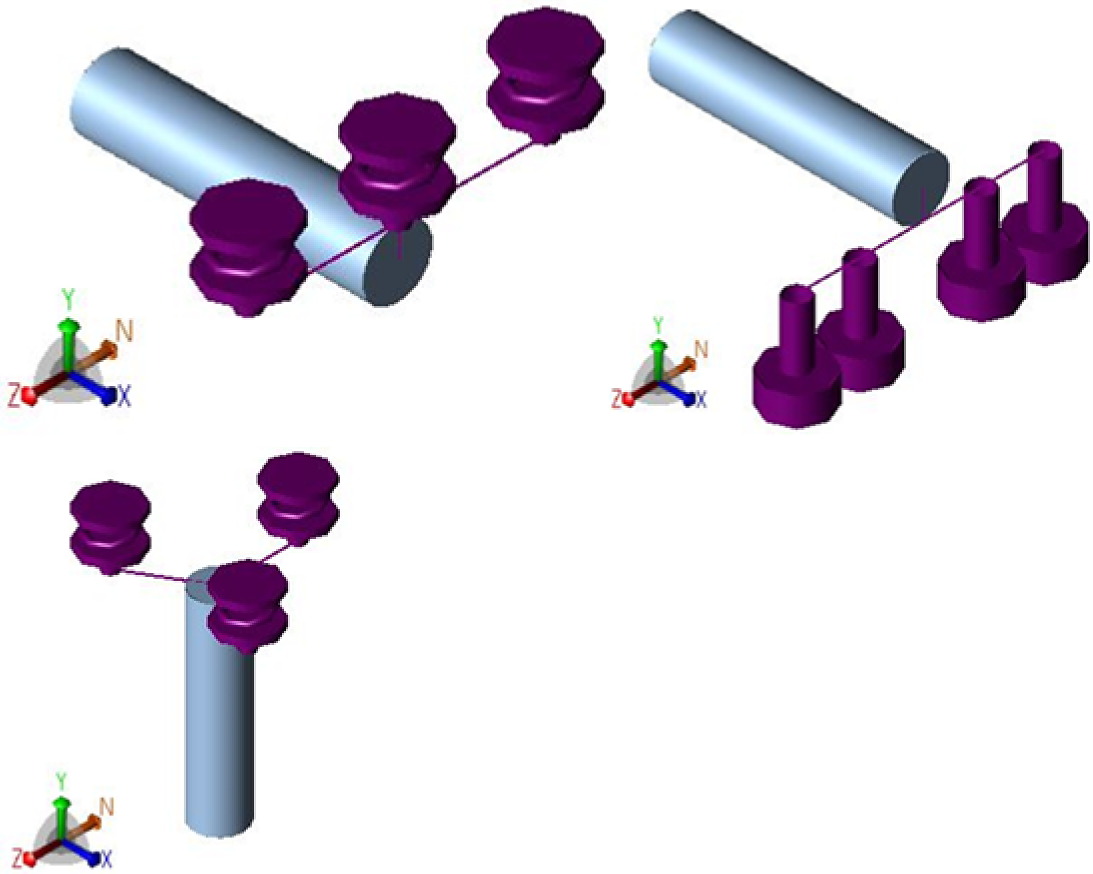
- A single hanger at the location of the hanger.



- A single can at the location of the can.



- Multiple hangers or cans at a single location as separate symbols.



Nozzle Flexibility

Piping Input menu: **Options** > **Nozzle Flexibility**

Piping Input Plot Tools toolbar: **Nozzles**

Static Output Processor menu: **PlotOptions** > **Nozzles**

Static Output Processor Plot Tools toolbar: **Nozzles**

Displays the nozzles for specifying stiffnesses.

Flange Check

Piping Input menu: **Options** > **Flange Check**

Piping Input Plot Tools toolbar: **Flanges**

Static Output Processor menu: **PlotOptions** > **Flange Check**

Static Output Processor Plot Tools toolbar: **Flanges**

Displays the flange nodes that the software evaluates.

Nozzle Check

Piping Input menu: **Options** > **Nozzle Check**

Piping Input Plot Tools toolbar: **Nozzle Limits**

Static Output Processor menu: **PlotOptions** > **Nozzle Check**

Static Output Processor Plot Tools toolbar: **Nozzle Limits**

Displays the nozzles to set a check.

Forces

Piping Input menu: **Options** > **Forces**

Piping Input Plot Tools toolbar: **Forces**

Static Output Processor menu: **PlotOptions** > **Forces**

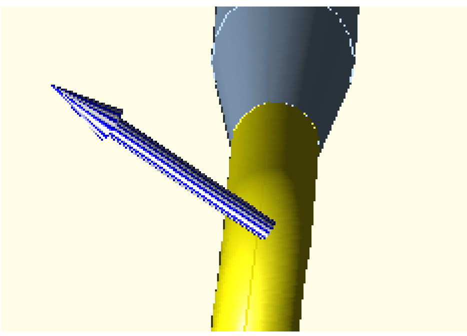
Static Output Processor Plot Tools toolbar: **Forces/Moments**

Shortcut key: **CTRL+ALT+F**

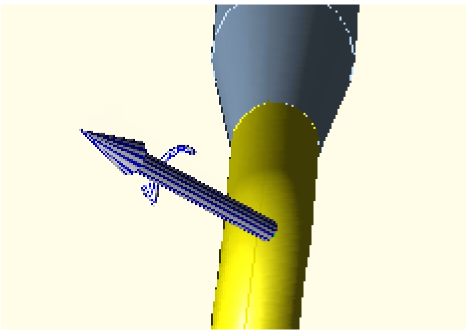
Turns the display of forces and moments on and off.

When **Forces** are turned on, the software displays:

- A directional arrow for a force.



- A directional arrow and curved arrow (following the right-hand rule) for a moment.



Other specifications:

- The arrow size - **Largest**, **Larger**, **Medium**, **Smaller**, or **Smallest**.
- The vector to display - **Vector1**, **Vector2**, and so on.

NOTE Change the default arrow colors in the Graphic Settings of the Configuration Editor. See Forces/Moments 1 and Forces/Moments 2.

Uniform Loads

Piping Input menu: **Options** > **Uniform Loads**

Piping Input Legends toolbar: **Uniform Loads**

Static Output Processor menu: **PlotOptions** > **Uniform Loads**

Static Output Processor Plot Tools toolbar: **Forces/Moments**

Updates the model to show each uniform load in a different color. Use this option to see the uniform load variations throughout the system or to verify that changes have been made. A color key displays the uniform loads defined in the model. You can change the assigned colors to meet your needs.

The uniform load parameters display in a table. Use the scroll bars to view all the data. Select **Next >>** and **Previous <<** to move through the displacement or force vectors.

Uniform Loads has three vectors defined. The **Node** column represents the start node number where the uniform loads vector was first defined. Because the data propagates throughout the model until changed or disabled, the model is colored accordingly.

Wind/Wave

Piping Input menu: **Options** > **Wind/Wave**

Legends toolbar: **Wind/Wave**

Static Output Processor menu: **PlotOptions** > **Wind/Wave**

Static Output Processor Legends toolbar: **Wind/Wave**

Updates the model to show each wind or wave load in a different color. Use this option to see the variations throughout the system or to verify that changes have been made. A color key displays the wind or wave loads defined in the model.

The wind and wave load parameters display in a table. Use the scroll bars to view all the data. Select **Next >>** and **Previous <<** to move through the loads.

All the elements with wind defined display in red. All the elements with wave data defined display in green. The legend grid shows the relevant data.

Axis

Piping Input menu: **Options** > **Axis**

Plot Tools toolbar: **Axis**

Static Output Processor menu: **PlotOptions** > **Axis**

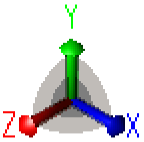
Static Output Processor Plot Tools toolbar: **Axis**

Shortcut key: **ALT**+**SHIFT**+**P**

Turns the display of the coordinate system on or off. Select the following options (in the Classic Piping Input Dialog and the Static Output Processor):

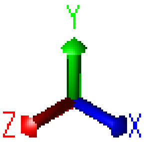
Axis Planes

Displays planes with the coordinate system axes.



Axis

Displays the coordinate system axes.



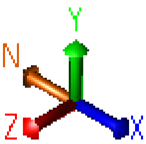
Off

Turns off axis display.

NOTE The software sets the default value from the value of Axis Mode in the Configuration Editor.

North Arrow

Displays a north arrow with the coordinate system. The North arrow indicates the North orientation of the plant.



NOTE The software sets the default value from the value of North Direction in the Configuration Editor.

Node Numbers

Piping Input menu: **Options** > **Node Numbers**

Piping Input Plot Tools toolbar: **Node Numbers**

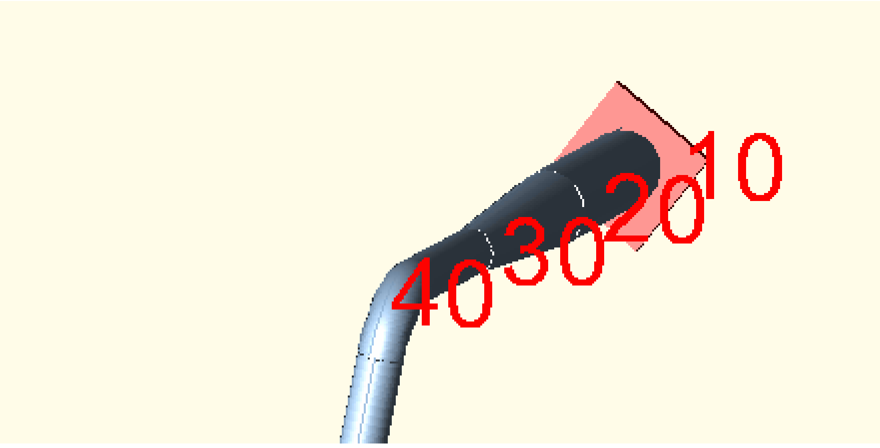
Static Output Processor menu: **PlotOptions** > **Node Numbers**

Static Output Processor Plot Tools toolbar: **Node Numbers**

Shortcut key: **CTRL**+**ALT**+**N**

Turns the display of node numbers on or off.

When node numbers are turned on, the software always displays the number in front of the pipe:



Also select and select **Node Display Options** to control the node number display:

Filters

Show All

Displays all node numbers or names.

Anchors

Displays anchor node numbers or names.

Hangers

Displays hanger node numbers or names.

Restraints

Displays restraint node numbers or names.

Format

Show Tags

Displays support tags, hanger tags, and element names.

Number Only

Displays node numbers if they are assigned. Node names do not display.

Name Only

Displays node names if they are assigned. Node numbers do not display.

The following formats display node name and node number:

- **Number (Name)**
- **Name (Number)**
- **Number - Name**
- **Name - Number**

Filter and Format Combinations

Select a format in combination with the **Show All**, **Anchors**, **Hangers**, or **Restraints** filters.

Customize node number, node name, and tag display by combining options, such as:

- **All** + **Show Tags** displays all node numbers, names, and tags.
- **Anchors** + **Show Tags** displays anchor node numbers, names, and tags.
- **All** + **Names Only** displays all node names. Node numbers and tags do not display.
- **Hangers** + **Names Only** displays hanger node names. Node numbers and tags do not display.
- **All** + **Show Tags** + **Names Only** displays all node names. Node numbers and tags do not display.

NOTE In cases where a node contains multiple values, a tag overrides a node name, and a node name overrides a node number.

Length

Piping Input menu: **Options** > **Length**

Plot Tools toolbar: **Lengths**

Static Output Processor menu: **PlotOptions** > **Length**

Static Output Processor Plot Tools toolbar: **Lengths**

Shortcut key: **ALT**+**SHIFT**+**L**

Turns the display of element lengths on or off. Alternatively, press **L**.

Tees

Piping Input menu: **Options** > **Tees**

Plot Tools toolbar: **Tees**

Static Output Processor menu: **PlotOptions** > **Tees**

Static Output Processor Plot Tools toolbar: **Tees**

Shortcut key: **ALT**+**SHIFT**+**T**

Displays where you have specified tees or SIFs on the model.

Expansion Joints & Rigids

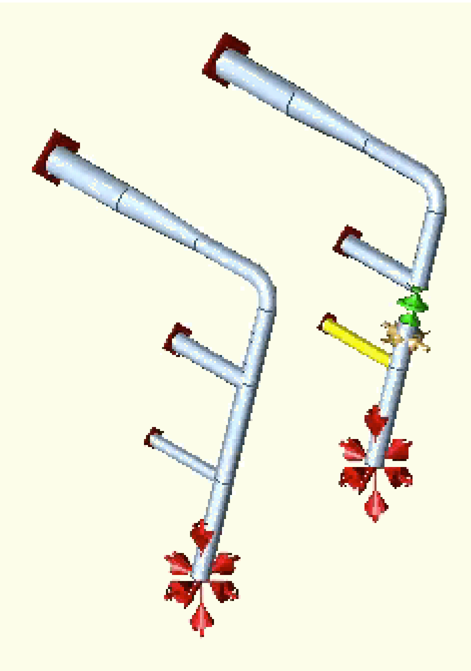
Piping Input menu: **Options** > **Expansion Joints**

Plot Tools toolbar: **Expansion Joints & Rigids**

Static Output Processor menu: **PlotOptions** > **Expansion Joints**

Static Output Processor Plot Tools toolbar: **Expansion Joints**

Emphasizes the display of elements such as restraints, anchors, displacements, and expansion joints by changing pipe element display to lines.



Diameters

Piping Input menu: **Options** > **Diameters**

Legends toolbar: **Diameters**

Static Output Processor menu: **PlotOptions** > **Diameters**

Static Output Processor Legends toolbar: **Diameters**

Shortcut key: **CTRL**+**ALT**+**D**

Updates the model to show each diameter in a different color. Use this option to see the diameter variations throughout the system or to verify that diameter changes have been made. Alternatively, press **D**. A color key displays the diameters defined in the model. Change the assigned colors to meet your needs and update diameter settings.

Wall Thickness

Piping Input menu: **Options** > **Wall Thicknesses**

Legends toolbar: **Wall Thicknesses**

Static Output Processor menu: **PlotOptions** > **Wall Thicknesses**

Static Output Processor Legends toolbar: **Wall Thicknesses**

Shortcut key: **CTRL+ALT+W**

Updates the model to show each wall thickness in a different color. Use this option to see the wall thickness variations throughout the system or to verify that changes have been made. Alternatively, press **W**. A color key displays the thicknesses defined in the model. Change the assigned colors to meet your needs.

Corrosion

Piping Input menu: **Options** > **Corrosion**

Legends toolbar: **Corrosion**

Static Output Processor menu: **PlotOptions** > **Corrosion**

Static Output Processor Legends toolbar: **Corrosion**

Updates the model to show each corrosion allowance in a different color. Use this option to see the corrosion variations throughout the system or to verify that changes have been made. A color key displays the corrosion allowances defined in the model. Change the assigned colors to meet your needs.

Piping Codes

Piping Input menu: **Options** > **Piping Codes**

Legends toolbar: **Piping Codes**

Static Output Processor menu: **PlotOptions** > **Piping Codes**

Static Output Processor Legends toolbar: **Piping Codes**

Updates the model to show each piping code in a different color. Use this option to see the piping code variations throughout the system or to verify that changes have been made.

Materials

Piping Input menu: **Options** > **Material**

Legends toolbar: **Materials**

Static Output Processor menu: **PlotOptions** > **Material**

Static Output Processor Legends toolbar: **Materials**

Shortcut key: **CTRL**+**ALT**+**M**

Updates the model to show each material in a different color. Use this option to see the material variations throughout the system or to verify that changes have been made. Alternatively, press **M**. A color key displays the materials defined in the model. Change the assigned colors to meet your needs.

Pipe Density

Piping Input menu: **Options** > **Pipe Density**

Legends toolbar: **Pipe Density**

Static Output Processor menu: **PlotOptions** > **Pipe Density**

Static Output Processor Legends toolbar: **Pipe Density**

Updates the model to show each pipe density in a different color. Use this option to see the pipe density variations throughout the system or to verify that changes have been made. A color key displays the pipe densities defined in the model. Change the assigned colors to meet your needs.

Fluid Density

Piping Input menu: **Options** > **Fluid Density**

Legends toolbar: **Fluid Density**

Static Output Processor menu: **PlotOptions** > **Fluid Density**

Static Output Processor Legends toolbar: **Fluid Density**

Updates the model to show each fluid density in a different color. Use this option to see the fluid density variations throughout the system or to verify that changes have been made. A color key displays the fluid densities defined in the model. Change the assigned colors to meet your needs.

Refractory Thickness

Piping Input menu: **Options** > **Refractory Thickness**

Static Output Processor menu: **PlotOptions** > **Refractory Thickness**

Updates the model to show each refractory thickness in a different color. Use this option to see the refractory thickness variations throughout the system or to verify that changes have been made. A color key displays the thicknesses defined in the model. Change the assigned colors to meet your needs.

Refractory Density

Piping Input menu: **Options** > **Refractory Density**

Static Output Processor menu: **PlotOptions** > **Refractory Density**

Updates the model to show each refractory density in a different color. Use this option to see the refractory density variations throughout the system or to verify that changes have been made. A color key displays the refractory densities defined in the model. Change the assigned colors to meet your needs.

Insulation Thickness

Piping Input menu: **Options** > **Insulation Thickness**

Legends toolbar: **Insulation**

Static Output Processor menu: **PlotOptions** > **Insulation Thickness**

Static Output Processor Legends toolbar: **Insulation**

Shortcut key: **ALT+SHIFT+I**

Updates the model to show each insulation thickness in a different color. Use this option to see the insulation thickness variations throughout the system or to verify that changes have been made. Alternatively, press **I**. A color key displays the thicknesses defined in the model. Change the assigned colors to meet your needs.

Insulation Density

Piping Input menu: **Options** > **Insulation Density**

Legends toolbar: **Insulation Density**

Static Output Processor menu: **PlotOptions** > **Insulation Density**

Static Output Processor Legends toolbar: **Insulation Density**

Updates the model to show each insulation density in a different color. Use this option to see the insulation density variations throughout the system or to verify that changes have been made. A color key displays the insulation densities defined in the model. Change the assigned colors to meet your needs.

Cladding Thickness

Piping Input menu: **Options** > **Cladding Thickness**

Static Output Processor menu: **PlotOptions** > **Cladding Thickness**

Updates the model to show each cladding thickness in a different color. Use this option to see the cladding thickness variations throughout the system or to verify that changes have been made. A color key displays the thicknesses defined in the model. Change the assigned colors to meet your needs.

Cladding Density

Piping Input menu: **Options** > **Cladding Density**

Static Output Processor menu: **PlotOptions** > **Cladding Density**

Updates the model to show each cladding density in a different color. Use this option to see the cladding density variations throughout the system or to verify that changes have been made. A color key displays the cladding densities defined in the model. Change the assigned colors to meet your needs.

Insul/Cladding Unit Wt

Piping Input menu: **Options** > **Insul/Cladding Unit Wt**

Static Output Processor menu: **PlotOptions** > **Insul/Cladding Unit Wt**

Updates the model to show each insulation or cladding unit weight in a different color. Use this option to see the variations throughout the system or to verify that changes have been made. A color key displays the insulation or cladding unit weights defined in the model. Change the assigned colors to meet your needs.

Temperatures

Piping Input menu: **Options** > **Temperatures** > **T1** ...**T9**

Piping Input Legends toolbar: **Show Temps** > **T1** ...**T9**

Static Output Processor menu: **Plot Options** > **Temperatures** > **T1** ...**T9**

Static Output Processor Legends toolbar: **Show Temps** > **T1** ...**T9**

Shortcut keys:

- **CTRL+1** - Show **Temperature 1**
- **CTRL+2** - Show **Temperature 2**
- CTRL+**3** - Show **Temperature 3**
- CTRL+**4** - Show **Temperature 4**
- CTRL+**5** - Show **Temperature 5**
- CTRL+**6** - Show **Temperature 6**
- CTRL+**7** - Show **Temperature 7**
- CTRL+**8** - Show **Temperature 8**
- CTRL+**9** - Show **Temperature 9**

Displays the temperature parameters that you have defined. Define up to nine temperature parameters.

Pressures

Piping Input menu: **Options** > **Pressures**

Piping Input Legends toolbar: **Show Pressures**

Static Output Processor menu: **PlotOptions** > **Pressures**

Static Output Processor Legends toolbar: **Show Pressures**

Displays the pressure parameters that you have defined.

Plot View Menu

Performs actions associated with viewing the model. **Plot View** commands area available after you select **Graphical Output**.

Reset

Piping Input menu: **View** > **Reset**

Piping Input Reset and Refresh Tools toolbar: **Reset Plot**

Static Output Processor menu: **PlotView** > **Reset**

Static Output Processor Reset Tools toolbar: **Restore/Reset**

Resets the view to the default settings. If a list has focus, resets the list to the default size.

Front View

Piping Input menu: **View** > **Front View**

Piping Input Standard Views toolbar: **Front**

Static Output Processor menu: **PlotView** > **Front View**

Static Output Processor Standard Views toolbar: **Front**

Graphic View right-click menu: **Views** > **Front**

Shortcut Key: **ALT+Z**

Displays the model from the front.

Back View

Piping Input menu: **View** > **Back View**

Piping Input Standard Views toolbar: **Back**

Static Output Processor menu: **PlotView** > **Back View**

Static Output Processor Standard Views toolbar: **Back**

Graphic View right-click menu: **Views** > **Back**

Shortcut key: **ALT**+**SHIFT**+**Z**

Displays the model from the back.

Top View

Piping Input menu: **View** > **Top View**

Piping Input Standard Views toolbar: **Top**

Static Output Processor menu: **PlotView** > **Top View**

Static Output Processor Standard Views toolbar: **Top**

Graphic View right-click menu: **Views** > **Top**

Shortcut key: **ALT**+**SHIFT**+**Y**

Displays the model from the top.

Bottom View

Piping Input menu: **View** > **Bottom View**

Piping Input Standard Views toolbar: **Bottom**

Static Output Processor menu: **PlotView** > **Bottom View**

Static Output Processor Standard Views toolbar: **Bottom**

Graphic View right-click menu: **Views** > **Bottom**

Shortcut key: **ALT**+**Y**

Displays the model from the bottom.

Left-side View

Piping Input menu: **View** > **Left-side View**

Piping Input Standard Views toolbar: **Left**

Static Output Processor menu: **PlotView** > **Left-side View**

Static Output Processor Standard Views toolbar: **Left**

Graphic View right-click menu: **Views** > **Left**

Shortcut key: **ALT**+**X**

Displays the model from the left side.

Right-side View

Piping Input menu: **View** > **Right-side View**

Piping Input Standard Views toolbar: **Right**

Static Output Processor menu: **PlotView** > **Right-side View**

Static Output Processor Standard Views toolbar: **Right**

Graphic View right-click menu: **Views** > **Right**

Shortcut key: **ALT**+**SHIFT**+**X**

Displays the model from the right side.

Southeast ISO View

Piping Input menu: **View** > **Southeast ISO View**

Piping Input Standard Views toolbar: **Southeast Isometric View**

Static Output Processor menu: **PlotView** > **Southeast ISO View**

Static Output Processor Standard Views toolbar: **Southeast Isometric View**

Graphic View right-click menu: **Views** > **SE Isometric**

Shortcut key: **F10**

Displays the model isometrically from the southeast.

Southwest ISO View

Piping Input menu: **View** > **Southwest ISO View**

Piping Input Standard Views toolbar: **Southwest Isometric View**

Static Output Processor menu: **PlotView** > **Southwest ISO View**

Static Output Processor Standard Views toolbar: **Southwest Isometric View**

Graphic View right-click menu: **Views** > **SW Isometric**

Displays the model isometrically from the southwest.

Northeast ISO View

Piping Input menu: **View** > **Northeast ISO View**

Piping Input Standard Views toolbar: **Northeast Isometric View**

Static Output Processor menu: **PlotView** > **Northeast ISO View**

Static Output Processor Standard Views toolbar: **Northeast Isometric View**

Graphic View right-click menu: **Views** > **NE Isometric**

Displays the model isometrically from the northeast.

Northwest ISO View

Piping Input menu: **View** > **Northwest ISO View**

Piping Input Standard Views toolbar: **Northwest Isometric View**

Static Output Processor menu: **PlotView** > **Northwest ISO View**

Static Output Processor Standard Views toolbar: **Northwest Isometric View**

Graphic View right-click menu: **Views** > **NE Isometric**

Displays the model isometrically from the northwest.

4 View

Piping Input menu: **View** > **4 View**

Piping Input Plot Tools toolbar: **4 View**

Static Output Processor menu: **PlotView** > **4 View**

Static Output Processor Plot Tools toolbar: **4 View**

Displays the model in four windows.

Drag the splitter bars to change the size of the windows. Drag the splitter bars out of the view to remove those views. Drag the splitter located at the top or left scroll bar to add views.

Individually manipulate the image in any of these panes.

Element Viewer Dialog

Use options in the **Element Viewer** dialog to navigate among the elements, navigate to various reports within a load case, and view the reports for other load cases. This is done in the **Report Selection** pane on the left.

The dialog has a tree structure similar in operation to Windows Explorer.

- Select the **+** sign for a load case to expand the tree to show reports.
- Select the report to display the data in the grid view to the right.
- Select a node or an element in the grid view when **Select Elements** is enabled to highlight the corresponding element on the graphics view.
- Zoom to the selected element if the corresponding **Zoom to Selection** is enabled. Similarly, select an element on the graphics view to highlight the corresponding data row in the report view. This is a bi-

directional connection.

- Change the load case within the **Element Viewer** dialog to update the graphics view (if applicable), and the **Load Case Selection** field on the **Load Case** toolbar.